

Memo NO. _____

Date / /

数学分析B1

X-Talk

1. 2011-2012 第一学期 第一次测试

1. $\forall \varepsilon > 0, \exists n_0 = [\frac{1}{4\varepsilon} + \frac{1}{2}] + 1, \forall n > n_0, \text{有}$

$$|\frac{n}{2n+\sin n} - \frac{1}{2}| = \frac{\frac{1}{2}|\sin n|}{2n+\sin n} = \frac{1}{2(\frac{2n}{|\sin n|} + 1)} \leq \frac{1}{2(2n-1)} < \varepsilon$$

2. ~~$\exists n_0$~~ $\exists X_0 > 0, \forall x_1, x_2 > X_0, \text{有 } |f(x_1) - f(x_2)| < \varepsilon$
 $\forall \varepsilon > 0,$

$$3. (1) = \lim_{n \rightarrow \infty} n^{\frac{3}{2}} \cdot \frac{\sin \frac{1}{n}}{\sqrt{n+\sin \frac{1}{n}} + \sqrt{n}} = \lim_{n \rightarrow \infty} n^{\frac{3}{2}} \cdot \frac{\frac{1}{n}}{2\sqrt{n}} = \frac{1}{2}$$

$$(2) = \lim_{x \rightarrow 1} \frac{x^2(x-1)+2(x-1)}{(x+1)(x-1)} = \lim_{x \rightarrow 1} \frac{x^2+2}{x+1} = \frac{3}{2}$$

$$(3) = \lim_{n \rightarrow \infty} [2^n \cdot (1 + (\frac{3}{2})^n)]^{\frac{1}{n}} = 2 \lim_{n \rightarrow \infty} [1 + (\frac{3}{2})^n]^{\frac{1}{n}} = 3$$

$$\Delta: x > 1 \text{ 时 } \lim_{n \rightarrow \infty} (1+x^n)^{\frac{1}{n}} = x:$$

$$(\text{省略 lim}) (1+x^n)^{\frac{1}{n}} = \exp \frac{\ln(1+x^n)}{n} = \exp \frac{\ln x^n + \ln(x^{-n}+1)}{n} = \exp \frac{\ln x^n}{n} = x$$

$$x < 1 \text{ 时 } \lim_{n \rightarrow \infty} (1+x^n)^{\frac{1}{n}} = 1:$$

$$1 = 1^{\frac{1}{n}} < (1+x^n)^{\frac{1}{n}} < 2^{\frac{1}{n}} \rightarrow 1$$

$$(4) = \lim_{x \rightarrow 0} (1 + \frac{1}{2}x^2)^{\frac{1}{x^2}} = \sqrt{\lim_{x \rightarrow 0} (1 + \frac{x^2}{2})^{\frac{2}{x^2}}} = \sqrt{e}$$

$$4. a_n \leq b^{n-1} a_1$$

$$\forall \varepsilon > 0, \exists n_0 = \dots, \forall n_1, n_2 > n_0, \text{有 } (S_{n_1} - S_{n_2}) < \varepsilon$$

$$S_{n_1} - S_{n_2} = \sum_{k=n_1+1}^{n_2} a_k \leq$$

$$S_n \uparrow \text{ (单调)}$$

$$S_n \leq \sum_{k=1}^n b^{k-1} a_1 = a_1 \cdot \frac{1-b^n}{1-b} \leq \frac{a_1}{1-b} \text{ (有界)} \Rightarrow \text{收敛}$$

Memo NO. _____

Date / /

~~先用数学归纳法证~~ $a_n \in (0, 1]$

$$5. a_{n+1} < \frac{a_n + a_n}{2} = a_n$$

$$\therefore a_n \downarrow \text{ (单调)} \\ a_n > 0 \text{ (有界)} \Rightarrow \text{收敛}$$

$$\text{设 } \lim_{n \rightarrow \infty} a_n = A$$

$$\therefore A = \frac{A + \sin A}{2} \Rightarrow A = \sin A \Rightarrow A = 0$$

$$6. x_n = \frac{x_{n-1} + a}{2} > \frac{x_{n-1} + x_{n-1}}{2} = x_{n-1} \text{ (单调)} \\ < \frac{a+a}{2} = a \text{ (有界)}$$

$$\lim_{n \rightarrow \infty} x_n = x_0 \Rightarrow x_0 = \frac{x_0 + a}{2} \Rightarrow x_0 = a$$

$$27. (1) \text{ 令 } y=0, |x f(x)| \leq M|x| \Rightarrow |f(x)| \leq M \Rightarrow \text{有界}$$

$$\therefore \lim_{x \rightarrow \infty} \frac{f(x)}{x} = 0$$

$$(2) \text{ 令 } a=0 \text{ 即可}$$

(题目有误, 应把条件改为 $|y f(x) - x f(y)| \leq M|x| + M|y|$):

$$(1) \left| \frac{f(x)}{x} - \frac{f(y)}{y} \right| \leq M \left| \frac{1}{y} \right| + M \left| \frac{1}{x} \right|$$

$$\forall \varepsilon > 0, \exists X_0 = \frac{2M}{\varepsilon}, \forall |x|, |y| > X_0, \text{有}$$

$$\left| \frac{f(x)}{x} - \frac{f(y)}{y} \right| \leq M \left(\left| \frac{1}{x} \right| + \left| \frac{1}{y} \right| \right) < \frac{2M}{X_0} = \varepsilon$$

$\Rightarrow$ 柯西收敛

$$(2) \text{ 设 } a = \lim_{x \rightarrow \infty} \frac{f(x)}{x}, \text{ 则 } \forall \varepsilon > 0, \left| \frac{f(x)}{x} - a \right| < \varepsilon$$

$$\left| f(x) - \frac{x}{y} f(y) \right| \leq M \frac{|x|}{|y|} + M$$

$$\text{令 } y \rightarrow \infty, \text{ 设 } a = \lim_{y \rightarrow \infty} \frac{f(y)}{y}, \text{ 则 } |f(x) - ax| \leq M \text{ (保号性)}$$

2011-2012 第一学期 第二次测试

1. (1) $f'(x) = e^{-x^2} - 2x^2 e^{-x^2} = e^{-x^2}(1-2x^2)$

令 $f'(x) = 0 \Rightarrow x = \pm \frac{1}{\sqrt{2}}$

$f''(x) = -2xe^{-x^2}(1-2x^2) + e^{-x^2} \cdot (-4x)$
 $= 2xe^{-x^2}(2x^2-3)$

令 $f''(x) = 0 \Rightarrow x = 0, \pm \sqrt{\frac{3}{2}}$

$f(\frac{1}{\sqrt{2}}) = \frac{1}{\sqrt{2}e}, f(-\frac{1}{\sqrt{2}}) = -\frac{1}{\sqrt{2}e}$

当 $x \rightarrow +\infty, f(x) = 0$, 当 $x \rightarrow -\infty, f(x) = 0$

$\therefore f(x)_{\max} = \frac{1}{\sqrt{2}e}, f(x)_{\min} = -\frac{1}{\sqrt{2}e}$

凹: $(-\infty, -\sqrt{\frac{3}{2}}), (0, \sqrt{\frac{3}{2}})$

凸: $(-\sqrt{\frac{3}{2}}, 0), (\sqrt{\frac{3}{2}}, +\infty)$

(2) $= \lim_{x \rightarrow +\infty} [(1+\frac{1}{x})^x] e^{-x} = 1$

(3) $= \lim_{x \rightarrow 0} \frac{2(1-\frac{x^2}{2}) - (1+3x) - \frac{9}{2}x^2 - (1-3x+\frac{9x^2}{2})}{-x^2} = \lim_{x \rightarrow 0} \frac{-x^2}{-x^2} = 1$

2. $f'(x) = \prod_{i=1}^k (x-x_i)^{n_i-1} \cdot \sum_{i=1}^k n_i \prod_{\substack{j=1 \\ (j \neq i)}}^k (x-x_j)^{n_j-1}$

$\therefore \sum_{i=1}^n n_i \prod_{\substack{j=1 \\ (j \neq i)}}^k (x-x_j)^{n_j-1} = \sum_{i=1}^k n_i \prod_{j=1}^{k-1} (x-x_j)^{n_j-1}$

$f(x) = 0 \Rightarrow \exists \xi_i, f'(\xi_i) = 0, \xi_i \in (x_i, x_{i+1})$

又 $g(\xi_i) = 0 \dots$

3. $y = \ln x$ 是凹函数
 $\therefore \ln(\frac{x_1^p}{p} + \frac{x_2^p}{q}) \geq \frac{1}{p} \ln(x_1^p) + \frac{1}{q} \ln(x_2^p) = \ln(x_1 x_2)$
 $\therefore x_1 x_2 \leq \frac{x_1^p}{p} + \frac{x_2^p}{q}$

4. $\frac{df'}{dx} = -f$

$G(x) = f^2 + f'^2$

$\Rightarrow f' \frac{df'}{dx} = -f \frac{df}{dx}$

$G'(x) = 0$

$\Rightarrow G(x) = C^2$

$\Rightarrow f' df = -f df$

$f' = \pm \sqrt{C^2 - f^2}$

$\Rightarrow \frac{1}{2} f'^2 = -\frac{1}{2} f^2 + C$

$\Rightarrow f^2 + f'^2 = 2C$

设 $f = A \sin \theta$, 则 $f' = A \cos \theta \cdot \frac{d\theta}{dx} = A \cos \theta$

$\therefore \frac{d\theta}{dx} = 1$

$\therefore f = A \sin(x + \varphi) \mid f' = A \cos(x + \varphi) = A \cos x \cos \varphi - A \sin x \sin \varphi$
 $= A \sin x \cos \varphi + A \cos x \sin \varphi$

而 $f(0) = A \sin \varphi, f'(0) = A \cos \varphi$

$\therefore f(x) = f(0) \cos x + f'(0) \sin x$

5. (1) 不妨设 $f'(a) > 0$, 则 $f'(b) > 0$

$\therefore \exists \delta > 0$, 使 $f(a+\delta) = f(a) + \delta f'(a+\varepsilon) > 0$ (保号性)

$f(b-\delta) = f(b) - \delta f'(b-\varepsilon) < 0$ (保号性)

$\therefore$ 由零点定理, $\exists \xi \in (a+\delta, b-\delta), f(\xi) = 0$

(2) $g(x) \triangleq f(x) e^{-x}, g'(x) = [f'(x) - f(x)] e^{-x}$

$\therefore g(a) = g(b) = g(\xi) = 0$

$\therefore \exists \xi \in (a, b), g'(\xi) = 0$ 即 $f'(\xi) = f(\xi)$

$\exists \xi_1 \in (a, \xi), \xi_2 \in (\xi, b), g'(\xi_1) = g'(\xi_2) = 0$

(3) $h(x) \triangleq [f(x) - f'(x)] e^{+x}, h'(x) = [f(x) - f''(x)] e^x$

$\therefore h(\xi_1) = h(\xi_2) = 0$

$\therefore \exists \eta \in (\xi_1, \xi_2), h'(\eta) = 0$

Memo NO. _____

Date / /

X-Talk

X-Talk

3. 2012-2013 第一学期 第一次测试

1. (1) $x \quad a_n = (-1)^n, a = 1$

(2) $\checkmark \quad \cancel{a_n = a_{n+1} = 0} \quad \forall \varepsilon > 0, \exists N > 0, \forall n_1 > N, n_2 > N, |a_{n_1} - a_{n_2}| < \frac{\varepsilon}{2}$
 $|a_{n_2} - a_{n_1}| < \frac{\varepsilon}{2} \Rightarrow |a_{n_1} - a_{n_2}| < \varepsilon \xrightarrow{\text{柯西收敛}} \text{收敛}$

(3) $\checkmark \quad g(x) = f(x) - x$
 $g(-2) \geq -1 - (-2) = 1 > 0, g(2) \leq 1 - 2 = -1 < 0$
 $\therefore \exists x_0 \in [-2, 2], f(x_0) = x_0$

(4) $\times \quad f(x) = x$

2. (1) = 1

(2) $= \lim_{n \rightarrow \infty} \left(n - \frac{(n+1)^{3/2}}{\sqrt{n}} \right) = \lim_{n \rightarrow \infty} n \left[1 - \left(1 + \frac{1}{n} \right)^{3/2} \right] = \lim_{n \rightarrow \infty} n \left[1 - \left(1 + \frac{3}{2n} \right) \right] = -\frac{3}{2}$

(3) $= \lim_{x \rightarrow +\infty} \left[\left(1 + \frac{2}{x+1} \right)^{\frac{x+1}{2}} \right]^4 / \left(1 + \frac{2}{x+1} \right)^2 = e^4$

(4) $= \lim_{x \rightarrow 0} \frac{\tan x \cdot \frac{1}{2} \sin x}{\frac{1}{2} \sin^2 x} = \lim_{x \rightarrow 0} \frac{1}{\cos x} = 1$

3. 当 $x > 0$ 时 $f(x) = \lim_{nx \rightarrow \infty} \frac{1+x^2 e^{nx}}{x+e^{nx}} = x^2$

当 $x = 0$ 时 $f(x) = 1$

当 $x < 0$ 时 $f(x) = \lim_{nx \rightarrow -\infty} \frac{1+x^2 e^{nx}}{x+e^{nx}} = \frac{1}{x}$

$\therefore f(x) = \begin{cases} x^2, & x > 0 \\ 1, & x = 0 \\ \frac{1}{x}, & x < 0 \end{cases}$

而 $\lim_{x \rightarrow 0^+} f(x) = 0, \lim_{x \rightarrow 0^-} f(x) = -\infty$

$\therefore f(x)$ 在 $x=0$ 处不连续, 其他地方连续

Memo NO. _____

Date / /

4. $f(x) = f(x^{\frac{1}{2}}) = f(x^{\frac{1}{4}}) = f(x^{\frac{1}{2^n}})$

$\therefore f(x)$ 在 $x=1$ 处连续

$\therefore \lim_{n \rightarrow \infty} f(x^{\frac{1}{2^n}}) = f(\lim_{n \rightarrow \infty} x^{\frac{1}{2^n}}) = f(1)$

$\therefore f(x) \equiv f(1)$ 为常数

5. $x_{n+1} = \frac{\alpha + \alpha x_n}{\alpha + x_n} > \frac{\alpha + x_n}{\alpha + x_n} = 1$ (用数学归纳法 $\forall n \geq 2, x_n > 1$)

$x_{n+1} = \frac{\alpha + \alpha x_n}{\alpha + x_n} < \frac{\alpha^2 + \alpha x_n}{\alpha + x_n} = \alpha$ (用数学归纳法 $\forall n \geq 2, x_n < \alpha$) (有界)

$x_{n+1} - x_n = \frac{\alpha + \alpha x_n - \alpha x_n - x_n^2}{\alpha + x_n} = \alpha - x_n > 0$

$\therefore \{x_n\}$ (单调)

$\therefore$ 有极限

$\therefore$ 设 $\lim_{n \rightarrow \infty} x_n = x_0, x_0 = \frac{\alpha(1+x_0)}{\alpha+x_0}$

$\therefore x_0 = \sqrt{\alpha}$

6. $\forall \varepsilon > 0, \exists N > \frac{1}{\varepsilon}, \forall n_1 > N, n_2 > N, \text{有}$ (不妨设 $n_2 > n_1$)

$|a_{n_1} - a_{n_2}| = \left| \sum_{k=n_1+1}^{n_2} \frac{(-1)^{k-1}}{k} \right|$
 若 $n_2 - n_1$ 为奇: $\sum_{k=n_1+1}^{n_2} \frac{(-1)^{k-n_1-1}}{k} < \frac{n_2 - n_1}{n_1} < \frac{1}{n_1} < \frac{1}{n_2}$

$= \begin{cases} \text{若 } n_2 - n_1 \text{ 为奇:} \\ \text{若 } n_2 - n_1 \text{ 为偶:} \end{cases} \quad S_{2n} \text{ 单调有界}$

交错级数审敛法(莱布尼兹定理) $\Rightarrow$ 收敛

7. $\therefore \lim_{n \rightarrow \infty} \frac{a_n}{n} = 0 \quad \forall k > N \quad \therefore \forall \varepsilon > 0, \exists N > 0, \forall k > N, \text{有 } |a_k| < \varepsilon |k| \leq \varepsilon n$

$\therefore \forall \varepsilon > 0, \exists N > 0, \forall n > N, \frac{|a_n|}{n} < \varepsilon$
 $M = \max_{i=1, \dots, N_0} \{a_i\}, \forall n > \max\{N_0, [\frac{M}{\varepsilon} + 1]\}, \forall i \leq n, \frac{|a_i|}{n} \leq \frac{M}{n} < \varepsilon$

$\therefore \frac{|a_n|}{n} < \varepsilon$

而 $\forall k \leq N, |a_k| < \varepsilon n$ 显然成立

$\therefore \frac{1}{n} \max_{1 \leq k \leq n} \{a_k\} < \frac{1}{n} \cdot \varepsilon n = \varepsilon \quad \therefore \lim_{n \rightarrow \infty} \frac{1}{n} \max_{1 \leq k \leq n} \{a_k\} = 0$

Memo NO. _____

Date / /

X-Talk

4. 2012-2013 第一学期 第二次测试

1. (1) $(x^2 e^x)^n = x^2 e^x + 2n x e^x + \frac{n(n-1)}{2} e^x = e^x (x^2 + 2n x + n^2 - n)$

(2) $\cos(xy) (y + x \frac{dy}{dx}) + 2y \frac{dy}{dx} = 1$

$$(\cos(xy) \cdot x + 2y) \frac{dy}{dx} = 1 - \cos(xy) \cdot y$$

$$\frac{dy}{dx} = \frac{1 - y \cdot \cos(xy)}{x \cdot \cos(xy) + 2y}$$

(3) $= \lim_{x \rightarrow 0} \frac{\sin x^2}{x^2 \sin x^2} = \infty$

(4) $= \lim_{x \rightarrow \infty} x a^{\frac{1}{x}} (1 - (\frac{b}{a})^{\frac{1}{x}}) = \lim_{t \rightarrow 0} \frac{a^t - b^t}{t} = \lim_{t \rightarrow 0} a^t \ln a - b^t \ln b$

(5) $= e^{\lim_{x \rightarrow 0} \frac{1}{x} \ln \frac{(1+x)^{\frac{1}{x}}}{e}} = \exp \left[\lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{x} \ln(1+x) - 1 \right) \right]$

$$= \exp \left[\lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{x} \cdot (x - \frac{x^2}{2}) - 1 \right) \right] = \exp(-\frac{1}{2}) = \frac{1}{\sqrt{e}}$$

(6) $= \lim_{x \rightarrow 0} \frac{\sin x \cos 2x \cos 3x + 2 \cos x \sin 2x \cos 3x + 3 \cos x \cos 2x \sin 3x}{\sin x}$

$$= \lim_{x \rightarrow 0} (\cos 2x \cos 3x + 4 \cos x \cos 3x + 3 \cos x \cos 2x (3 - 4 \sin^2 x))$$

$$= 1 + 4 + 9 = 14$$

(7) $\rho = \frac{(1+y'^2)^{3/2}}{|y''|} = \frac{(1+(\frac{1}{x})^2)^{3/2}}{|-\frac{1}{x^2}|} = x^2 [1 + \frac{1}{x^2}]^{3/2}$

$$\frac{1}{\rho} = \frac{1}{x^2 (1 + \frac{1}{x^2})^{3/2}}$$

2. 令 $f(x) = x^{\frac{1}{x}}, x \in [1, +\infty)$

$$f'(x) = x^{\frac{1}{x}} \cdot (-\frac{1}{x^2} \ln x + \frac{1}{x^2}) = x^{\frac{1}{x}-2} (1 - \ln x)$$

$\therefore (1, e) \uparrow, (e, +\infty) \downarrow$

而 $f(e) = e^{\frac{1}{e}} \therefore \sqrt[n]{n} \leq e^{\frac{1}{e}}, \forall n \in \mathbb{N}^+$

Memo NO. _____

Date / /

X-Talk

$x: \sqrt[3]{2} < \sqrt[3]{3}$

$\therefore A = \sqrt[3]{3}, n_0 = 3$

3. ~~$x > 0$ 时, $f'(x)$~~

只需 $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x)$ 且 $f'_+(x) = f'_-(x)$

$\lim_{x \rightarrow 0^+} f(x) = 0, \lim_{x \rightarrow 0^-} f(x) = b \Rightarrow b = 0$

$f'_+(x) = \lim_{x \rightarrow 0^+} \frac{\tan^2(ax)}{x^2} = a^2, f'_-(x) = \lim_{x \rightarrow 0^-} \frac{(2a-1)x+b}{x} = 2a-1 \Rightarrow a=1$

4. 充分性: $\forall x_1, x_2 \in I, \frac{|f(x_1) - f(x_2)|}{|x_1 - x_2|} = |f'(\xi)| < M, \xi \in (x_1, x_2)$

(设 $x_1 < x_2$)

$\therefore \forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{M}, \forall |x_1 - x_2| < \delta, \text{ 有 } |f(x_1) - f(x_2)| < \delta M = \varepsilon$

必要性不成立: eg. $f(x) = \sqrt{x}, I = [0, 1]$

5. 假设无界, 即 $\forall M > 0, \exists x_0, \text{ 使 } |f(x_0)| > M$

不妨设 $M \geq f(a), M \geq f(b)$. 令 $M = \max\{|f(a)|, |f(b)|, |f(\frac{a+b}{2})|\}$

令 $x = a, \lambda x + (1-\lambda)y = x_0$, 调整 λ , 可使 y 取遍 (x_0, b)

则 $f(\lambda a + (1-\lambda)b) \leq \lambda f(a) + (1-\lambda)f(b) \leq M$

$\therefore f(x) \leq M, \forall x \in [a, b]$

又取 $x = a, \lambda x + (1-\lambda)y = \frac{a+b}{2}$, 调整 λ , 可使 y 取遍 $(\frac{a+b}{2}, b)$

则 $f(\frac{a+b}{2}) \leq \lambda f(a) + (1-\lambda)f(y)$

$$-M \leq \lambda M + (1-\lambda)f(y)$$

$f(y) \geq -\frac{1+\lambda}{1-\lambda} M \geq -3M, \forall y \in (\frac{a+b}{2}, b)$

同理 $f(y) \geq -3M, \forall y \in (a, \frac{a+b}{2})$ 因区间凸函数 $\Rightarrow$ 用区间连续 $\Rightarrow$ 有界

$\therefore \forall x \in [a, b], \text{ 有 } |f(x)| \leq 3M$

6. $\frac{f(x) - f(0)}{x - 0} = \lim_{\Delta x \rightarrow 0} \frac{f(0x + \Delta x) - f(0x)}{\Delta x}$

(导) $f'(x) = \theta f''(\theta x) x + f'(\theta x)$

(降) $\frac{f'(x)}{x} = \theta f''(\theta x) + \frac{f'(\theta x)}{x}$

(洛) $f''(x) = \theta f''(\theta x) + \theta f''(\theta x)$

$$\therefore \theta = \frac{1}{2}$$

(课本惠民P197有另解)

Memo NO. _____

Date / /

X-Talk

5. 2013-2014 第一学期 第一次测试

1. $\forall \varepsilon > 0, \exists N = \left\lceil \frac{1}{2\varepsilon} \right\rceil + 1, \forall n > N$, 有

$$\left| \frac{n}{2n+(-1)^n \sqrt{2}} - \frac{1}{2} \right| = \left| \frac{(-1)^n \sqrt{2}}{2(2n+(-1)^n \sqrt{2})} \right| \leq \frac{\sqrt{2}}{2n} < \varepsilon$$

2. (1) $\times$ ~~$\exists \varepsilon = \frac{1}{2}, \forall N > 0, \exists n > N$~~ 取 $a_n = 2n-1, b_n = 2n$.

有 $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = +\infty$. 但 $\lim_{n \rightarrow \infty} f(a_n) = -1, \lim_{n \rightarrow \infty} f(b_n) = 1$.

$\therefore$ 不收敛

(2) $\checkmark$ $u_k = \frac{\ln k}{k^3}, u_{k+1} < u_k$

交错级数审敛法 $\Rightarrow$ 收敛

3. (1) $\sum_{k=1}^n \frac{1}{n+(-1)^k \sqrt{k}} > \sum_{k=1}^n \frac{1}{n+\sqrt{k}} > \sum_{k=1}^n \frac{1}{n+\sqrt{n}} = \frac{n}{n+\sqrt{n}}$

而 $\lim_{n \rightarrow \infty} \frac{n}{n+\sqrt{n}} = 1 \Rightarrow$ 原式 ≥ 1

同理, 原式 ≤ 1

$\therefore$ 原式 $= 1$

(2) $= \lim_{x \rightarrow 0} (1+2x)^{\frac{1}{2x}} = \sqrt[2x]{1+2x} = \sqrt{e}$

(3) $\lim_{x \rightarrow 0} \frac{\frac{1}{x} + 1}{x} = \frac{1}{n}$

(4) $= \lim_{n \rightarrow \infty} n^\alpha \left[\left(1 + \frac{\ln n}{n} \right)^\alpha - 1 \right] = \lim_{n \rightarrow \infty} n^\alpha \cdot \frac{\alpha \ln n}{n} = \lim_{n \rightarrow \infty} \frac{\alpha \ln n}{n^{1-\alpha}}$
 $= \lim_{x \rightarrow +\infty} \frac{\alpha \ln x}{x^{1-\alpha}} = \lim_{x \rightarrow +\infty} \frac{\alpha/x}{(1-\alpha)x^{-\alpha}} = 0$

4. $A = \{f(x) | a < x < b\}$ 有上确界 β .

$\forall \varepsilon > 0, \exists x_1 \in (a, x_0)$, 有 $\beta - \varepsilon < f(x_1) \leq \beta$. 取 $\delta = x_0 - x_1 > 0$

$\forall 0 < x_0 - x < \delta$, 有 $x_1 < x < x_0$, $\beta - \varepsilon < f(x_1) < f(x) \leq \beta$ 即 $|f(x) - \beta| < \varepsilon$

$\therefore f(x_0^-) = \beta$ 存在, 同理 $f(x_0^+)$ 也存在 (以上 f 改为 g)

单侧连续的复合 $\Rightarrow f(x_0^-), f(x_0^+)$ 存在

Memo NO. _____

Date / /

X-Talk

5. $\forall \varepsilon > 0, \exists N > 0, \forall n > N$, 有 $|a_n - a| < \varepsilon$

$$|a_{n_1} - a| < \varepsilon, |a_{n_2} - a| < \varepsilon \Rightarrow |a_{n_1} - a_{n_2}| \leq |a_{n_1} - a| + |a_{n_2} - a| < 2\varepsilon$$

取 $\varepsilon = \frac{1}{2}$, 则 $a_{n_1} = a_{n_2}$, 即 $\forall n > N$, 有 $a_n = C \Rightarrow |a_n - C| < \varepsilon$

$$\therefore \lim_{n \rightarrow \infty} a_n = C \Rightarrow a = C$$

$$\therefore \exists N > 0, \forall n > N, \text{有 } a_n = a$$

6. (1) $a_1 \geq 1$

② 假设 $a_k \geq 1$, 则 $a_{k+1} = 1 + \frac{2}{a_k} \geq 1$

$$\therefore a_n \geq 1, \forall n \Rightarrow a_n = 1 + \frac{2}{a_{n-1}} \leq 3$$

$$a_{n+1} - a_n = 1 + \frac{2}{a_n} - a_n = \frac{-a_n^2 + a_n + 2}{a_n} = \frac{-(a_n - 2)(a_n + 1)}{a_n}$$

$$a_{n+1} - a_n = \frac{2}{a_n} - \frac{2}{a_{n-1}} = \frac{2(a_{n-1} - a_n)}{a_n a_{n-1}} \text{ (保留)} \Rightarrow a_{n+1} - a_n \text{ 与 } a_{n-1} - a_{n-2} \text{ 同号}$$

$$a_n - a_{n-2} = \frac{2}{a_{n-1}} - \frac{2}{a_{n-3}} = \frac{2(a_{n-3} - a_{n-1})}{a_{n-1} a_{n-3}}$$

$$a_{2k-1} < a_{2k}, a_{2k} > a_{2k+1}$$

$\therefore a_n - a_{n-2}$ 与 $a_{n-3} - a_{n-1}$ 同号

$\therefore a_n - a_{n-2}$ 与 $a_{n-2} - a_{n-4}$ 同号

而 $a_1 = 1, a_2 = 3, a_3 = \frac{5}{3}, a_4 = \frac{11}{5} \Rightarrow a_1 < a_3, a_2 > a_4$

$$\therefore a_{2k-1} < a_{2k+1}, a_{2k} > a_{2k+2}$$

$\therefore \{a_{2k-1}\} \uparrow, \{a_{2k}\} \downarrow$ 单调有界 $\Rightarrow$ 收敛

$$\because a_{2k-1} < a_{2k}$$

$$\therefore \text{(保号性)} \lim_{k \rightarrow \infty} a_{2k-1} \leq \lim_{k \rightarrow \infty} a_{2k}$$

$$\because a_{2k} > a_{2k+1} \therefore \text{(保号性)} \lim_{k \rightarrow \infty} a_{2k} \geq \lim_{k \rightarrow \infty} a_{2k+1} = \lim_{k \rightarrow \infty} a_{2k-1} \left. \begin{array}{l} \lim_{k \rightarrow \infty} a_{2k-1} = \lim_{k \rightarrow \infty} a_{2k} \\ = \lim_{n \rightarrow \infty} a_n \end{array} \right\}$$

(2) $a = 1 + \frac{2}{a} \Rightarrow \lim_{n \rightarrow \infty} a_n = a = 2$

另解: $a_{2n+1} = 3 - \frac{4}{a_{2n+2}}$

归纳得 $a_{2k+1} < a_{2k+3} < 2$ 单调有界

6. 2013-2014 第一学期 第二次测试

1. $|x| > 1$ 时, $f(x) = \frac{1}{x}$

$$\left. \begin{aligned} f(1) &= \frac{1+a+b}{2} = \lim_{x \rightarrow 1^+} f(x) = 1 \Rightarrow a+b=1 \\ f(-1) &= \frac{-1+a-b}{2} = \lim_{x \rightarrow 1^-} f(x) = -1 \Rightarrow a-b=-1 \end{aligned} \right\} \rightarrow a=0, b=1$$

验证: $|x| < 1$ 时, $f(x) = x^a$, 成立

$$\therefore a=1, b=0 \quad \therefore a=0, b=1$$

$$2. f(0) = \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{f(x)}{x} \cdot \lim_{x \rightarrow 0} x = 0$$

$$f'(0) = \lim_{x \rightarrow 0} \frac{f'(x)}{1} = \lim_{x \rightarrow 0} \frac{f(x)-f(0)}{x-0} = 0$$

$$\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = \lim_{x \rightarrow 0} \frac{f'(x)}{2x} = \lim_{x \rightarrow 0} \frac{f''(x)}{2} = \frac{f''(0)}{2}$$

$$3. \lim_{x \rightarrow x_0} \frac{(x-x_0)f'(x_0) - [f(x)-f(x_0)]}{[f(x)-f(x_0)](x-x_0)f'(x_0)} = \lim_{x \rightarrow x_0} \frac{-\frac{1}{2}(x-x_0)^2 f''(x_0)}{[f(x)-f(x_0)](x-x_0)f'(x_0)}$$

$$= \lim_{x \rightarrow x_0} \frac{-\frac{1}{2}f''(x_0)}{f'(x_0)f'(\frac{1}{2})} = -\frac{f''(x_0)}{2[f'(x_0)]^2}$$

$$4. \frac{dy}{dx} = \frac{y'}{x'} = \frac{e^t(\cos t - \sin t)}{e^t(\sin t + \cos t)} = \frac{1 - \tan t}{\tan t + 1}$$

$$\frac{d^2y}{dx^2} = \frac{-\sec^2 x (\tan t + 1) - (1 - \tan t) \sec^2 x}{(\tan t + 1)^2 e^t (\sin t + \cos t)} = -\frac{2\sec^2 x}{(\tan t + 1)^2 e^t (\sin t + \cos t)}$$

$$5. g(x) \triangleq |f(x)|$$

若存在 $x=0$ 的邻域内 $f(x)$ 同号, 则 $\begin{cases} f(0) \geq 0: g'(0) = f'(0) \\ f(0) < 0: g'(0) = -f'(0) \end{cases}$

否则 $f(0)=0$. 不妨设 $x \in (-\delta, 0)$ 时, $f(x) < 0$; $x \in (0, \delta)$ 时, $f(x) > 0$.

$$\text{则 } g'_+(0) = f'_+(0) = f'(0), g'_-(0) = -f'_-(0) = -f'(0)$$

要使 $g'(0) = g'_+(0) = g'_-(0)$, 则必有 $f'(0)=0$

$$6. \text{令 } x=y, f(x^2) = 2f(x)$$

$$\text{令 } x=y=1, \begin{cases} f(1) = 2f(1) \Rightarrow f(1)=0 \\ f(x) = 2f(x^{\frac{1}{2}}) = 2^n f(x^{\frac{1}{2^n}}) = \lim_{n \rightarrow \infty} 2^n f(x^{\frac{1}{2^n}}) \end{cases}$$

$$\lim_{y \rightarrow 1} f(xy) = f(x) + \lim_{y \rightarrow 1} f(y) = f(x)$$

$$\lim_{x \rightarrow x_0} f(x) \xrightarrow{\text{连续}}$$

$$7. \Leftarrow \frac{\sqrt{n+1}}{2\ln n} \ln \sqrt{n} > \sqrt{n} \ln \sqrt{n+1}$$

$$\Leftarrow \frac{2\ln n}{\sqrt{n}} > \frac{2\ln(n+1)}{\sqrt{n+1}}$$

$$\Leftarrow \frac{\ln n}{\sqrt{n}} > \frac{\ln(n+1)}{\sqrt{n+1}}$$

$$f(x) \triangleq \frac{\ln x}{\sqrt{x}}, f'(x) = \frac{\frac{1}{x} - \frac{\ln x}{2\sqrt{x}}}{x} = \frac{2 - \ln x}{2x\sqrt{x}} < 0 \quad (x \geq 10 \text{ 时})$$

$\therefore f(x)$ 在 $[10, +\infty) \downarrow$

$\therefore f(n) > f(n+1)$

8. $\int_0^9 v(t) dt = 100$ 设 $x(t)$ 表示第 $t \sim t+1$ 秒跑过的路程.

若 $\forall t \in [0, 9], x(t) > 10$, 则 $\sum_{t=0}^9 x(t) > 100. \times$

若 $\forall t \in [0, 9], x(t) < 10$, 则 $\sum_{t=0}^9 x(t) < 100. \times$

若 $\exists t_0 \in [0, 9], x(t_0) = 10$, 则已成立

若 $\exists t_1, t_2 \in [0, 9], x(t_1) < 10, x(t_2) > 10$, 则由介值定理,

$\exists t_0 \in (t_1, t_2)$ 或 $t_0 \in (t_2, t_1)$, 使 $x(t_0) = 10$.

9. 若 $\exists f(t) \neq 0$

$\Rightarrow \exists$ 最大值 $f(x_0) > 0$ (最小值同理)

$f'(x_0) = 0, f''(x_0) \leq 0$ 与 $f'(x_0) = e^{x_0} f(x_0) > 0$ 矛盾

Memo NO. _____

X-Talk

Date / /

7 2015-2016 第一学期 第一次测试

$$1. = \lim_{x \rightarrow 1} \frac{x-1}{(x-1)(x+1)^2} = \lim_{x \rightarrow 1} \frac{1}{(x+1)^2} = \frac{1}{4} \text{ (不想用定义)}$$

$$2. = \frac{\sqrt{n}(-1)^n}{\sqrt{n+(-1)^n} + \sqrt{n}}$$

$$\lim_{n \rightarrow \infty} [\sqrt{n}(\sqrt{n+(-1)^n} - \sqrt{n})] = \lim_{n \rightarrow \infty} \frac{\sqrt{n}(-1)^n}{2\sqrt{n}} = \lim_{n \rightarrow \infty} \frac{(-1)^n}{2} \text{ 不收敛}$$

(2个子列分别收敛于 $\pm \frac{1}{2}$)

$$3. (1) = 0$$

$$(2) = \exp\left[\lim_{n \rightarrow \infty} n\left(1 + 2\sin\frac{1}{n}\right)\right] = \exp\left[\lim_{n \rightarrow \infty} n\left(1 + \frac{2}{n} + o\left(\frac{1}{n}\right)\right)\right] = \exp\left[\lim_{n \rightarrow \infty} n\left(\frac{2}{n} + o\left(\frac{1}{n}\right)\right)\right] = e^2$$

$$4. (1) = \lim_{x \rightarrow 1} \frac{(x+3)(x-1)}{2(x-1)(x+5)} = \lim_{x \rightarrow 1} \frac{x+3}{2(x+5)} = \frac{1}{3}$$

$$(2) = \lim_{x \rightarrow 0} \frac{1 - \frac{1}{2}x^2 - 1 + \frac{1}{2}(2x)^2}{x^2} = \frac{3}{2}$$

$$5. \text{无界. } \forall M > 0, \exists x_0 = \frac{\pi}{2}M, x_0 \sin x_0 = \frac{\pi}{2}M > M.$$

$$\text{不是无穷大量. } \exists M > 0, \forall x_0 > 0, \exists x = \pi x_0, |x \sin x| = 0 < M$$

$$6. \lim_{n \rightarrow \infty} 1 = \frac{1}{2}, \lim_{n \rightarrow \infty} \frac{1}{n} = -\frac{1}{2}$$

$$7. \lim_{n \rightarrow \infty} \frac{a_n^2}{n} = \lim_{n \rightarrow \infty} \frac{a_n^2 - a_{n-1}^2}{1} = \lim_{n \rightarrow \infty} \frac{a_n + a_{n-1}}{a_{n-1}} = \lim_{n \rightarrow \infty} \frac{2a_{n-1} + \frac{1}{a_{n-1}}}{a_{n-1}} = 2$$

$\therefore \lim_{n \rightarrow \infty} \frac{a_n}{\sqrt{n}} = \sqrt{2}$ ($\lim_{n \rightarrow \infty} \frac{1}{a_n} = 0$)

$$8. a_{n+1} - a_n = \frac{1}{2}\left(1 - \frac{1}{n}\right)(a_n - a_{n-1}) < \frac{1}{2}(a_n - a_{n-1}) < \dots < \frac{1}{2^n}$$

$$\therefore a_n < 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}} < 3$$

单调有界 $\Rightarrow$ 收敛

Memo NO. _____

Date / /

X-Talk

8 2015-2016 第一学期 第二次测试

$$1. (1) (cx+d)f(x) = ax+b$$

$$(cx+d)C_n^0 f^{(n)}(x) + C_n^1 c f^{(n-1)}(x) = 0, n \geq 2 \text{ 时}$$

$$\Rightarrow f^{(n)}(x) = -\frac{nc}{(cx+d)} f^{(n-1)}(x) = (-1)^{n-1} \frac{c^{n-1} n!}{(cx+d)^{n-1}} \left(\frac{ax+b}{cx+d}\right)' = (-1)^{n-1} \frac{c^{n-1} n!}{(cx+d)^{n-1}} \cdot \frac{ad-bc}{(cx+d)^2}$$

$$(2) x' = -\cos t, y' = \sin t$$

$$\frac{dy}{dx} = \frac{\sin t}{-\cos t}$$

$$\frac{d^2 y}{dx^2} = \frac{\cos t(1 - \cos t) - (\sin t)^2}{(-\cos t)^3} = -\frac{1}{(1 - \cos t)^2}$$

$$2. (1) = \lim_{x \rightarrow 0} \frac{1 - x \cot x}{x^2} = \lim_{x \rightarrow 0} \frac{-\cot x + x/\sin^2 x}{2x} = \lim_{x \rightarrow 0} \frac{-\cos x + \frac{x}{\sin x}}{2x \sin x}$$

$$= \lim_{x \rightarrow 0} \frac{-\frac{1}{2} \sin 2x + x}{2x^3} = \lim_{x \rightarrow 0} \frac{-\cos 2x + 1}{6x^2} = \lim_{x \rightarrow 0} \frac{2 \sin 2x}{12x} = \frac{1}{3}$$

$$(2) = \lim_{x \rightarrow +\infty} \left[\frac{e}{2}x + x^2 \left(e^{x \ln(1 + \frac{1}{x})} - e \right) \right]$$

$$= \lim_{x \rightarrow +\infty} \left[\frac{e}{2}x + x^2 \left(e^{x(\frac{1}{x} - \frac{1}{2x^2} + \frac{1}{3x^3})} - e \right) \right]$$

$$= \lim_{x \rightarrow +\infty} \left[\frac{e}{2}x + x^2 \left(e^{-\frac{1}{2x} + \frac{1}{3x^2}} - 1 \right) \right]$$

$$= \lim_{x \rightarrow +\infty} \left[\frac{e}{2}x + x^2 e \left(-\frac{1}{2x} + \frac{1}{3x^2} + \frac{1}{8x^3} \right) \right]$$

$$= \frac{11e}{24}$$

$$3. (1) f(t) = t^{\frac{1}{n}}, f'(t) = \frac{1}{n} t^{\frac{1}{n}-1}, f''(t) = \left(\frac{1}{n}-1\right) \frac{1}{n} t^{\frac{1}{n}-2} < 0$$

$$f(x) \triangleq x^{\frac{1}{n}} - y^{\frac{1}{n}} - (x-y)^{\frac{1}{n}}, x \geq y$$

$$f'(x) = \frac{1}{n} x^{\frac{1}{n}-1} - \frac{1}{n} (x-y)^{\frac{1}{n}-1} = \frac{1}{n} \left(\frac{1}{x^{1-\frac{1}{n}}} - \frac{1}{(x-y)^{1-\frac{1}{n}}} \right) \leq 0$$

$$\therefore f(x) \downarrow \Rightarrow f(x) \leq f(y) = 0$$

Memo NO. _____

Date / /

X-Talk

$$(2) f(x) = \cos x - 1 + \frac{x^2}{2}$$

$$f'(x) = -\sin x + x \quad (-\infty, 0) \downarrow, (0, +\infty) \uparrow \Rightarrow f(x) > f(0) = 0$$

$$f''(x) = 1 - \cos x \geq 0$$

$$4. f'(x) = -e^{-x} \sum_{k=0}^n \frac{x^k}{k!} + e^{-x} \sum_{k=1}^n \frac{x^{k-1}}{(k-1)!}$$

$$= -e^{-x} \cdot \frac{x^n}{n!}$$

$$\text{令 } f'(x) = 0 \Rightarrow x = 0$$

若 n 为奇, $f(x) = f(0) = 1$, 无极小值

若 n 为偶, $f'(x) \leq 0$, $f(x)$ 无极大、小值

5. $\therefore f(x)$ 一致连续

$$\therefore \forall \varepsilon > 0, \exists \delta > 0, \forall |x_1 - x_2| < \delta, \text{ s.t. } |f(x_1) - f(x_2)| < \varepsilon$$

$$\text{令 } h(y_1) = y_1^\alpha - y_2^\alpha - (y_1 - y_2)^\alpha, y_1 \geq y_2$$

$$h'(y_1) = \alpha y_1^{\alpha-1} - \alpha(y_1 - y_2)^{\alpha-1} \leq 0$$

$$\therefore h(y_1) \downarrow \Rightarrow h(y_1) \leq h(y_2) = 0$$

$$\therefore y_1^\alpha - y_2^\alpha \leq (y_1 - y_2)^\alpha$$

$$\therefore \forall \varepsilon > 0, \exists \delta > 0, \forall |x_1 - x_2| < \delta, \text{ s.t. } |f(x_1) - f(x_2)| < \varepsilon^{\frac{1}{\alpha}}$$

$$\Rightarrow |g(x_1) - g(x_2)| \leq |f(x_1) - f(x_2)|^\alpha < \varepsilon$$

$$6. p'(x) = \sum_{k=1}^n x^{k-1} = \sum_{k=0}^{n-1} x^k = \frac{1-x^n}{1-x}, x \neq 1$$

~~$x \neq 1$ 时, 若 n 是偶数, $(-\infty, -1) \downarrow, (-1, 1) \uparrow, (1, +\infty) \uparrow \Rightarrow p(x) \geq p(1)$~~

$p(x)$ 的零点就是 $f(x) = e^{-x} \sum_{k=0}^n \frac{x^k}{k!}$ 的零点.

由 4 知, n 为奇时, $f(x)$ 在 $(-\infty, 0) \uparrow, (0, +\infty) \downarrow$, 在 $(-\infty, 0)$ 有一零点.

n 为偶时, $f(x)$ 在 $\mathbb{R} \downarrow$, $\lim_{x \rightarrow -\infty} f(x) = +\infty$, $\lim_{x \rightarrow +\infty} f(x) = 0$, 无零点.

Memo NO. _____

Date / /

X-Talk

15 2019-2020 第一学期 期中考试

$$1. (1) 0 < a_n < M$$

$$\forall \varepsilon > 0, \exists n_0, \forall n > n_0, \text{ 有}$$

$$\frac{a_n}{a_1 + a_2 + \dots + a_n} < \varepsilon, \forall \varepsilon > 0,$$

① 若 $\exists \delta > 0$, 有无穷多项 $a_k > \delta$, 则 $\exists n_0$, 使 $a_i (i=1, \dots, n_0)$ 中有 $[\frac{M}{\varepsilon \delta}] + 1$

$$\text{项} > \delta, \text{ 则 } \frac{a_n}{a_1 + a_2 + \dots + a_n} < \frac{M}{[\frac{M}{\varepsilon \delta}] + 1} \cdot \delta \leq \varepsilon$$

$$\therefore \lim_{n \rightarrow \infty} \frac{a_n}{a_1 + a_2 + \dots + a_n} = 0$$

② 若 $\forall \delta > 0$, 仅有有限项 $a_k > \delta$, 则 $\exists n_0$, 使 $\forall n > n_0$, 有 $a_n \leq \delta$

$$\text{取 } \delta = \varepsilon a_1, \text{ 则 } \frac{a_n}{a_1 + a_2 + \dots + a_n} < \frac{\varepsilon a_1}{a_1} = \varepsilon$$

$$\therefore \lim_{n \rightarrow \infty} \frac{a_n}{a_1 + a_2 + \dots + a_n} = 0$$

$$(2) = \lim_{x \rightarrow +\infty} \left(x \sqrt{1 + \frac{3}{x}} + \frac{2}{x^2} + ax + b \right)$$

$$= \lim_{x \rightarrow +\infty} \left(x \left(1 + \frac{3}{2x} \right) + ax + b \right)$$

$$= \lim_{x \rightarrow +\infty} [(1+a)x + (\frac{3}{2} + b)] = 0 \Rightarrow a = -1, b = -\frac{3}{2}$$

$$\text{证 (3)} = \lim_{x \rightarrow x_0} \frac{(x-x_0)f'(x_0) - [f(x) - f(x_0)]}{[f(x) - f(x_0)](x-x_0)f'(x_0)}$$

$$= \lim_{x \rightarrow x_0} \frac{\frac{1}{2}f''(x_0)[f'(x_0)x - f(x)] - [f'(x_0)x_0 - f(x_0)]}{[f(x) - f(x_0)](x-x_0)f'(x_0)}$$

同 13

Memo NO. _____

Date / /

X-Talk

$$(4) \frac{dy}{dx} = \frac{2t}{1+t^2} = 2t$$

$$\frac{d^2y}{dx^2} = 2(1+t^2)$$

$$(5) f^{(n)}(0) = \frac{d^n}{dx^n} \left[\frac{2 \ln(1-x^2)}{1-x^2} \right]_{x=0} = 2 \left[\frac{-2x}{1-x^2} \right]_{x=0}^{(n-3)}$$

$$\text{设 } g(x) = \frac{2x}{x^2-1}, \text{ 则 } (x^2-1)g(x) = 2x$$

$$(x^2-1)g^{(k)}(x) + k \cdot 2x g^{(k-1)}(x) + k(k-1)g^{(k-2)}(x) = 0, k \geq 2 \text{ 时}$$

$$\text{令 } x=0, \text{ 得 } g^{(k)}(0) = k(k-1)g^{(k-2)}(0)$$

$$\because g(0)=0, g'(0)=2$$

$$\therefore g^{(k)}(0) = \begin{cases} 0, & k \text{ 为偶} \\ -2 \cdot k!, & k \text{ 为奇} \end{cases}$$

$$\therefore f^{(n)}(0) = \begin{cases} 0, & n \text{ 为奇} \\ -4 \cdot (n-3)!, & n \text{ 为偶} \end{cases}$$

$$(6) = \lim_{x \rightarrow 0} \frac{(1 - \frac{1}{2} \sin^2 x + \frac{1}{24} \sin^4 x) - (1 - \frac{1}{2} x^2 + \frac{1}{24} x^4)}{x^4}$$

$$= \lim_{x \rightarrow 0} \frac{1}{x^4} \left[-\frac{1}{2} (x - \frac{1}{6} x^3)^2 + \frac{1}{24} x^4 + \frac{1}{2} x^2 - \frac{1}{24} x^4 \right]$$

$$= \lim_{x \rightarrow 0} \frac{1}{x^4} \cdot \left(\frac{1}{6} x^4 \right) = \frac{1}{6}$$

$$\text{2. } g(x) = f'(x)$$

$$[f(x^2)]'' = f''(x^2) \cdot 2x \Rightarrow [f(x^2)]''_{x=0} = 0$$

Memo NO. _____

Date / /

X-Talk

$$3. \lim_{x \rightarrow 0} \frac{f(x)}{x} = x^{\alpha-1} \sin \frac{1}{x}$$

$$(1) \alpha \leq 1$$

$$(2) x \neq 0 \text{ 时, } f'(x) = \alpha x^{\alpha-1} \sin \frac{1}{x} + x^{\alpha} \cos \frac{1}{x} \cdot \left(-\frac{1}{x^2}\right) = x^{\alpha-1} \left(\alpha \sin \frac{1}{x} - \frac{1}{x} \cos \frac{1}{x} \right)$$

$$\lim_{x \rightarrow 0} f'(x) \neq f'(0) = 0 \Rightarrow 1 < \alpha \leq 2$$

$$(3) \alpha > 2$$

4. $\lim_{n \rightarrow \infty} f(x_n) = A$ 若 $\exists x_1, x_2 \in [a, b], f(x_1) < f(x_2)$, 则 $\exists x_0$ 介于 x_1, x_2 之间, $f(x_0) = A$.
否则, 不妨设 $f(x) \geq A$.

$$\therefore \forall \varepsilon > 0, \exists n_0 > 0, \forall n > n_0, |f(x_n) - A| < \varepsilon \Rightarrow A \leq f(x_n) < A + \varepsilon$$

$$\therefore \inf \{f(x_n)\} = A \Rightarrow \checkmark$$

(否则, 设 $\inf \{f(x_n)\} = m > A$, 取 $\varepsilon = m - A$, 则 $\forall n, f(x_n) \geq A + \varepsilon$ 矛盾)

$$\text{收敛子列 } \{x_{n_k}\}, f(x_0) = \lim_{k \rightarrow \infty} f(x_{n_k}) = \lim_{k \rightarrow \infty} f(x_{n_k}) = \lim_{n \rightarrow \infty} f(x_n) = A$$

$$5. (1) \forall \varepsilon > 0, \exists \delta = \max \{|f'(x)|\}, \forall x_1, x_2 \geq a \text{ 且 } |x_1 - x_2| < \frac{\varepsilon}{\delta}, \text{ 有}$$

$$|f(x_1) - f(x_2)| = |f'(\xi)(x_1 - x_2)| = |f'(\xi)| \cdot |x_1 - x_2| < \varepsilon$$

$$(2) \frac{|f(x)|}{x} \leq \frac{|f'(\xi)(x-a) + f(a)|}{x} = |f'(\xi)| \cdot \frac{|x-a|}{x} + \frac{|f(a)|}{a} < K \text{ (有界)}$$

$$\left| \frac{f(x_1)}{x_1} - \frac{f(x_2)}{x_2} \right| = \left| \frac{x_2 f(x_1) - x_1 f(x_2)}{x_1 x_2} \right| = \frac{|(x_2 - x_1)(f(x_1) - f(x_2)) + (x_2 f(x_2) - x_1 f(x_1))|}{x_1 x_2}$$

$$\leq \frac{(x_1 - x_2)^2 |f'(\xi)|}{x_1 x_2} + \frac{|f(\xi_2) + \xi_2 f'(\xi_2)| \cdot |x_1 - x_2|}{x_1 x_2}$$

$$\leq |x_1 - x_2| \cdot \left(\frac{|x_1 - x_2| \cdot M}{a^2} + \frac{|f(\xi_2)| + \xi_2 |f'(\xi_2)|}{a \xi_2} \right)$$

$$< \frac{\delta^2 M}{a^2} + \frac{\delta K}{a} + \frac{\delta M}{a} < \varepsilon \quad (\exists \delta)$$

$$6. (1) g(x) = e^{-x} f(x)$$

$$g'(x) = -e^{-x} (f(x) - f'(x)) < 0$$

$$g(x) \downarrow, g(x) < g(0) = 1 \Rightarrow f(x) < e^x$$

$$(2) g(x) = e^{-x} (f(x) + f'(x))$$

$$g'(x) = -e^{-x} (f(x) - f'(x)) < 0$$

$$g(x) < g(0) = 2 \Rightarrow f(x) + f'(x) \leq 2e^x$$

$$h(x) = e^x (f(x) - f'(x)) > h(0) = 0$$

$$h(x) = e^x (f(x) - e^x) < h(0) = 0$$

$$\therefore f(x) < e^x$$

16 2020-2021 第一学期期中考试

$$1. (1) \sqrt[n]{1 + \frac{1}{2} + \dots + \frac{1}{n}} < \sqrt[n]{1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \dots + \frac{1}{2^{\lfloor \log_2 n \rfloor}}} < \sqrt[n]{\lfloor \log_2 n \rfloor + 2}$$

$$< \sqrt[n]{\log_2 n + 3} < 1 + \varepsilon \rightarrow \sqrt[n]{n} = 1 + \lambda_n \rightarrow 0$$

$$\Leftrightarrow \log_2 n + 3 < (1 + \varepsilon)^n$$

$$\Leftrightarrow \log_2 n + 2 < n\varepsilon$$

$$\text{而 } \lim_{n \rightarrow \infty} \frac{\log_2 n + 2}{n} = 0$$

$$n = 1 + n\lambda_n + \frac{n(n-1)}{2}\lambda_n^2 > 1 + \frac{n(n-1)}{2}\lambda_n^2$$

$$\lambda_n < \sqrt{\frac{2}{n}}$$

$$\therefore \text{成立, 即 } \lim_{n \rightarrow \infty} \sqrt[n]{1 + \frac{1}{2} + \dots + \frac{1}{n}} = 1$$

$$(2) = \lim_{x \rightarrow \infty} \left(2e^{\frac{1}{x}} + \frac{\cos x}{x} \right) = 2 + 0 = 2$$

$$(3) = \lim_{x \rightarrow \infty} \frac{\frac{1}{2}x - \frac{1}{3}x}{\frac{1}{3}x} = 1$$

$$(4) f'(x) = (\sin x)^{\cos x} (\cos x \cdot \ln \sin x)' \\ = (\sin x)^{\cos x} \left(-\sin x \cdot \ln \sin x + \cos x \cdot \frac{1}{\sin x} \cdot \cos x \right) \\ = (\sin x)^{\cos x} \left(-\sin x \cdot \ln \sin x + \frac{\cos^2 x}{\sin x} \right)$$

$$(5) f(x) = \left(2 - \left(1 + \frac{1}{2}x^2 - \frac{1}{24}x^4 \right)^{\frac{1}{3}} \right) + o(x^5) \\ = \left(1 + \frac{1}{2}x^2 - \frac{1}{24}x^4 \right)^{\frac{1}{3}} + o(x^5) \\ = 1 + \frac{1}{3} \left(\frac{1}{2}x^2 - \frac{1}{24}x^4 \right) + \frac{\frac{1}{3} \cdot (-\frac{2}{3})}{2} \cdot \left(\frac{1}{2}x^2 - \frac{1}{24}x^4 \right)^2 + o(x^5) \\ = 1 + \frac{1}{6}x^2 - \frac{1}{72}x^4 - \frac{1}{9} \left(\frac{1}{4}x^4 \right) + o(x^5) \\ = 1 + \frac{1}{6}x^2 - \frac{1}{24}x^4 + o(x^5)$$

$$2. a_{n+1} = -(a_n - 1)^2 + 1 \leq 1 \quad (\text{有界})$$

归纳可知 $a_n > 0$

$$a_{n+1} - a_n = a_n(1 - a_n) \geq 0 \quad (\text{单调})$$

$\therefore \{a_n\}$ 收敛

$$\text{设 } \lim_{n \rightarrow \infty} a_n = a_0, \text{ 则 } a_0 = a_0(2 - a_0) \Rightarrow a_0 = 1$$

$$3. \lim_{x \rightarrow 0^+} f(x) = 0 \Rightarrow a = 2$$

$$\lim_{x \rightarrow 0^-} f(x) = 0 \Rightarrow b = 1$$

$$\text{此时, } \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{x^2 \sin \frac{1}{x}}{x} = \lim_{x \rightarrow 0^+} x \sin \frac{1}{x} = \lim_{x \rightarrow 0^+} \frac{\sin \frac{1}{x}}{\frac{1}{x}} = 1$$

$$f'(x) = \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{x \cos x}{x} = \lim_{x \rightarrow 0^+} \cos x = 1$$

$\therefore$ 可导

$$4. f(x) = x^2 - x \sin x - \cos x + \frac{1}{2}$$

$$f'(x) = 2x - x \cos x - \sin x + \sin x = (2 - \cos x)x$$

$$\therefore f(x) \text{ 在 } (-\infty, 0) \downarrow, (0, +\infty) \uparrow$$

$$\text{而 } \lim_{x \rightarrow -\infty} f(x) = +\infty, \lim_{x \rightarrow +\infty} f(x) = +\infty, f(0) = -\frac{1}{2} < 0$$

$$\therefore \text{ 恰有 } \exists x_1 \in (-\infty, 0), x_2 \in (0, +\infty), \text{ 使 } f(x_1, x_2) = 0$$

$$15. \forall \varepsilon > 0, \exists \delta > 0, \forall |x_1 - x_2| < \delta,$$

$$\textcircled{1} \text{ 若 } x_1, x_2 \in (a, b] \text{ 或 } x_1, x_2 \in [b, c), \text{ 则已成立}$$

$$\textcircled{2} \text{ 若 } \text{否} \text{ 则, 不妨设 } x_1 \in (a, b], x_2 \in [b, c). \text{ 取 } \delta = \min \left\{ \frac{b-a}{2}, \frac{c-b}{2} \right\}$$

$$\therefore \text{ 在 } (a, b] \text{ 和 } [b, c) \text{ 连续}$$

$$\therefore \text{ 在 } (a, c) \text{ 连续}$$

$$\therefore \text{ 在 } [b - \delta, b + \delta] \text{ 连续}$$

$$\therefore \text{ 在 } [b - \delta, b + \delta] \text{ 一致连续}$$

$$\therefore x_1, x_2 \in [b - \delta, b + \delta]$$

$$\therefore |f(x_1) - f(x_2)| < \varepsilon$$

$$\therefore f(x) \text{ 在 } (a, c) \text{ 上一致连续}$$

$$\leq |f(x_1) - f(b)| + |f(b) - f(x_2)|$$

6. $\frac{x dx}{a^2} + \frac{y dy}{b^2} = 0$

在 (x_0, y_0) ($x_0 > 0, y_0 > 0$) 处的切线为 $\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} = 1$

令 $x=0$, 得 $y = \frac{b^2}{y_0}$

令 $y=0$, 得 $x = \frac{a^2}{x_0}$

$\therefore S = \frac{1}{2} \cdot \frac{b^2}{y_0} \cdot \frac{a^2}{x_0} = \frac{ab}{2 \sin \theta \cos \theta} = \frac{ab}{\sin 2\theta}$

当且仅当 $\theta = \frac{\pi}{4}$ 时有 $S_{\min} = ab$

此时, $x_0 = a \sin \theta = \frac{\sqrt{2}}{2} a, y_0 = \frac{\sqrt{2}}{2} b$

切线: $\frac{\sqrt{2}x}{2a} + \frac{\sqrt{2}y}{2b} = 1$

7. 原式 $\Leftrightarrow \frac{\frac{f(b)}{b} - \frac{f(a)}{a}}{\frac{1}{b} - \frac{1}{a}} = f(\xi) - \xi f'(\xi)$

设 $g(x) = \frac{f(x)}{x}, h(x) = \frac{1}{x}$

由柯西中值定理, $\frac{g(b) - g(a)}{h(b) - h(a)} = \frac{g'(\xi)}{h'(\xi)} = \frac{\frac{f'(\xi)\xi - f(\xi)}{\xi^2}}{-\frac{1}{\xi^2}} = f(\xi) - \xi f'(\xi)$

$\forall \epsilon > 0, \lim_{x \rightarrow 0} \frac{f(x) - f(ax)}{x - ax} = \frac{A}{1-a}$

$\lim_{x \rightarrow 0} \frac{f(x) - f(ax)}{x} = \lim_{x \rightarrow 0} \left[\frac{f(x) - f(0)}{x-0} - \frac{f(ax) - f(0)}{x-0} \right]$
 $= \lim_{x \rightarrow 0} \left[\frac{f(x) - f(0)}{x-0} - a \cdot \frac{f(ax) - f(0)}{ax-0} \right]$
 $= f'(0) - a f'(0)$

$A' \triangleq \frac{A}{1-a}$

$A' - \frac{\epsilon}{2} < \frac{f(a^{n+1}x) - f(a^n x)}{a^{n+1}x - a^n x} < A' + \frac{\epsilon}{2}$

$A' - \frac{\epsilon}{2} < \frac{f(x) - f(a^n x)}{x - a^n x} < A' + \frac{\epsilon}{2}$

令 $n \rightarrow \infty, A' - \frac{\epsilon}{2} \leq \frac{f(x) - f(0)}{x-0} \leq A' + \frac{\epsilon}{2}$

13 2017-2018 第一学期期中考试

2. $f(x) = \sin \frac{1}{x}$

取 $\epsilon = 1, \forall \delta > 0, \exists x_1 = \frac{1}{2n\pi - \frac{\pi}{2}}, x_2 = \frac{1}{2n\pi + \frac{\pi}{2}} (n = [\max\{\frac{1}{2\delta}, 1\}] + 1)$

$|x_1 - x_2| = \frac{\pi}{(2n\pi)^2 - \frac{\pi^2}{4}} < \frac{\pi}{2n\pi} = \frac{1}{2n} < \delta$

而 $|f(x_1) - f(x_2)| = 2 > \epsilon \Rightarrow$ 不一致连续

3. (1) $= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1}\right)^{n+1} \cdot \frac{1}{1 - \frac{1}{n+1}}$

$= \frac{1}{e}$

(2) $= \lim_{x \rightarrow 0} \frac{x \cdot \frac{x}{2} \cdot x^2}{[(x - \frac{x^3}{6}) - x(1 - \frac{x^2}{2})]x} = \frac{3}{4}$

(3) $= \lim_{x \rightarrow 1^-} \frac{\ln(1-x)}{\frac{1}{\ln x}} = \lim_{x \rightarrow 1^-} \frac{\frac{1}{x-1}}{-\frac{1}{\ln^2 x} \cdot \frac{1}{x}} = \lim_{x \rightarrow 1^-} \frac{\ln^2 x}{1-x}$
 $= \lim_{x \rightarrow 1^-} \frac{2 \ln x \cdot \frac{1}{x}}{-1} = 0$

(4) $= \lim_{x \rightarrow \infty} (x - x^2(\frac{1}{x} - \frac{1}{2x^2})) = \frac{1}{2}$

4. $\sum_{k=1}^n \frac{k}{n^2+k} > \sum_{k=1}^n \frac{k}{n^2+n} = \frac{(1+n)n/2}{n^2+n} \rightarrow \frac{1}{2}, n \rightarrow \infty$ 时

$\sum_{k=1}^n \frac{k}{n^2+k} < \sum_{k=1}^n \frac{k}{n^2} = \frac{(1+n)n/2}{n^2} \rightarrow \frac{1}{2}, n \rightarrow \infty$ 时

$\therefore \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k}{n^2+k} = \frac{1}{2}$

5. 同 4) T3.

Memo NO. _____

Date / /

X-Talk

6. $y' + 2^y \ln 2 \cdot y' - 1 - \cos x = 0$

$x=0$ 时 $y+2^y=1 \Rightarrow y=0$

代入上式, 得 $y' + \ln 2 \cdot y' - 2 = 0 \Rightarrow y'(0) = \frac{2}{\ln 2 + 1}$

7. $g(x) = e^{-\frac{x}{b-a}} (f(x) - f(a))$, $g(a) = 0$

$g'(x) = -e^{-\frac{x}{b-a}} \left(\frac{f(x) - f(a)}{b-a} \right) \cdot \left(\frac{f(x) - f(a)}{b-a} - f'(x) \right)$

若 $\exists c \in [a, b]$, $f(c) = f(a) \Rightarrow g(c) = 0$, 则 $\exists \xi \in (a, c)$, $g'(\xi) = 0 \Rightarrow \checkmark$

否则, 不妨设 $\forall x \in [a, b]$, $f(x) > f(a)$, 则

$g'(x_0) = -e^{-\frac{x_0}{b-a}} \left(\frac{f(x_0) - f(a)}{b-a} - 0 \right) < 0$

~~$\times \frac{1}{x-a^+} f'_+(a) > 0 \Rightarrow f(a+\delta) > 0 (\exists \delta > 0)$~~

$\therefore g'(a+\delta)$

$g(x_0) > 0 \Rightarrow g'(\eta) = \frac{g(x_0) - g(a)}{x_0 - a} > 0 (\exists \eta \in (a, x_0))$

$\therefore \exists \xi \in (\eta, x_0)$, s.t. $g'(\xi) = 0$

Memo NO. _____

Date / /

X-Talk

[14] 2018-2019 第一学期 期中考试

2. (1) $= \lim_{n \rightarrow \infty} [\ln(n+1) + \ln n] \cdot [\ln(n+1) - \ln n]$

$= \lim_{n \rightarrow \infty} \ln[n(n+1)] \cdot \ln \frac{n+1}{n}$

$= \lim_{n \rightarrow \infty} [\ln(n+1) + \ln n] \cdot \ln \left(1 + \frac{1}{n}\right)$

$= \lim_{n \rightarrow \infty} \frac{\ln(n+1) + \ln n}{n}$

$= \lim_{n \rightarrow \infty} [\ln(n+1) - \ln(n-1)] = 0$

(2) $= \lim_{n \rightarrow \infty} n^{\frac{2}{3}} \cdot n^{\frac{1}{3}} \left[\left(1 + \frac{1}{n}\right)^{\frac{1}{3}} - 1 \right]$

$= \lim_{n \rightarrow \infty} n \cdot \frac{1}{3n} = \frac{1}{3}$

(3) $= \lim_{x \rightarrow +\infty} \left(1 - \frac{1}{x+2}\right)^{x+2} \cdot \left(1 - \frac{1}{x+2}\right)^{-2} = \frac{1}{e}$

(4) $= \lim_{x \rightarrow +\infty} -\frac{e}{2}$

3. (1) $= \frac{1}{\tan \frac{x}{2}} \cdot \sec^2 \frac{x}{2} \cdot \frac{1}{2} = \frac{1}{2 \sin \frac{x}{2} \cos \frac{x}{2}} = \frac{1}{\sin x}$

(2) $= \frac{1}{\sqrt{1 - \left(\frac{1-x^2}{1+x^2}\right)^2}} \cdot \left(-\frac{2}{(1+x^2)^2}\right) 2x = -\frac{2}{1+x^2}$

(3) $= \frac{1}{2\sqrt{1-x^2}} \cdot (-2x) = -\frac{x}{\sqrt{1-x^2}}$

(4) $= x e^x + n e^x$

4. $a_n \in [1, 2]$

$a_{n+1} - a_n = \frac{1}{a_n} - \frac{1}{a_{n-1}} = \frac{a_{n-1} - a_n}{a_n a_{n-1}}$

$\{a_{2k-1}\} \uparrow, \{a_{2k}\} \downarrow$ 同 5T6

6. $g(x) = xf(x) + x - 1$

$g(0) = -1 < 0, g(1) = f(1) \geq 0$

$\therefore \exists x_0 \in [0, 1], g(x_0) = 0 \Rightarrow f(x_0) = \frac{1-x_0}{x_0}$

设 $x_1 \neq x_2$ 满足 $f(x_1) = \frac{1-x_1}{x_1}, f(x_2) = \frac{1-x_2}{x_2}$

则 $f(x_1) - f(x_2) = \frac{x_2 - x_1}{x_1 x_2}$

$|f(x_1) - f(x_2)| = \frac{1}{x_1 x_2} |x_2 - x_1| > |x_2 - x_1|$, 不符

$\therefore x_0$ 唯一

7. $(f''(x))^2 \leq (f'(x))^2$

$[f'(x) - f''(x)] \cdot [f'(x) + f''(x)] \geq 0$

~~$e^x [f'(x) - f''(x)] \cdot e^{+x} [f'(x) + f''(x)] \geq 0$~~

~~$[e^{-x} f'(x)]' \cdot [e^x f'(x)]' \leq 0$~~

若不严格单调, 则 $\exists x_0, s.t. f'(x_0) = 0, f''(x_0) \neq 0$

代入上式, 得 $-[f''(x_0)]^2 \geq 0$, 矛盾

(反证法) 若不(严格单调), 则 $\exists x_0, s.t. f'(x_0) = 0$

设 $M = \max_{x \in [x_0 - \frac{1}{2}, x_0 + \frac{1}{2}]} |f'(x)| = |f'(x_m)|$

(微分中值定理) $\frac{M}{\frac{1}{2}} \leq \left| \frac{f'(x_m) - f'(x_0)}{x_m - x_0} \right| = |f''(\xi)| \leq |f'(\xi)| \leq M \Rightarrow M = 0$

$\therefore f'(x) \equiv 0, \forall x \in [x_0 - \frac{1}{2}, x_0 + \frac{1}{2}]$

一直往外拓, 得 $f'(x) \equiv 0, \forall x \in (-\infty, +\infty)$, 与“非常值”矛盾

12 2007-2008 第一学期期中考试

2. (1) $\sum_{k=1}^n \frac{1}{(n^3+n-k)^{\frac{1}{3}}} < \sum_{k=1}^n \frac{1}{n^{\frac{3}{3}}} = 1$

$\sum_{k=1}^n \frac{1}{(n^3+n-k)^{\frac{1}{3}}} > \sum_{k=1}^n \frac{1}{(n^3+n)^{\frac{1}{3}}} = \frac{1}{(1+\frac{1}{n^3})^{\frac{1}{3}}} \rightarrow 1$

$\therefore$ 原式 = 1

(2) $= \lim_{x \rightarrow 0} \frac{x^3 \sin^2 \frac{1}{x}}{\frac{1}{2} x^2} = \lim_{x \rightarrow 0} 2x \sin^2 \frac{1}{x} = 0$

(3) $= \lim_{x \rightarrow +\infty} \left(\left(1 + \frac{2}{x-1} \right)^{\frac{x-1}{2}} \right)^2 \cdot \left(1 + \frac{2}{x-1} \right) = e^2$

(4) $= \lim_{x \rightarrow 0} \frac{\frac{1}{6} x^3}{x^3} = \frac{1}{6}$

3. (1) $= 2^x \ln 2 \cdot \sin x + 2^x \cos x + \frac{2x}{x^2+1}$

4. ~~$f'(x)$~~ $f(x) = \frac{1}{2} \sin 2x (\sin x + \cos x)$
 $= \frac{\sqrt{2}}{2} \sin 2x \cdot \sin(x + \frac{\pi}{4})$

$= \frac{\sqrt{2}}{4} (\cos(3x + \frac{\pi}{4}) + \cos(x - \frac{\pi}{4}))$

$f'(x) = \frac{\sqrt{2}}{4} (3 \sin(3x + \frac{\pi}{4}) + \sin(x - \frac{\pi}{4}))$

$x \in [0, \frac{\pi}{4})$ 时, $f'(x) > 0$; $x = \frac{\pi}{4}$ 时, $f'(x) = 0$; $x \in (\frac{\pi}{4}, \frac{\pi}{2}]$ 时, $f'(x) < 0$

$\therefore f(x)_{\max} = \frac{f(\frac{\pi}{4})}{2} = \frac{\sqrt{2}}{2}$

5. $g(x) = e^{x^2} f(x), g(a) = g(b) = 0$

$g'(x) = e^{x^2} (2x f(x) - f'(x)) \leq 0$

若 $\exists c \in (a, b), s.t. f(c) > 0$ (<0 同理) $\Rightarrow g(c) > 0$

则 $\exists \xi \in (a, c), s.t. g'(\xi) = \frac{g(c) - g(a)}{c - a} > 0$, 矛盾

$\therefore f(x) \equiv 0, \forall x \in [a, b]$

Date / /

X-Talk

$$6. g(x) = A[f(x)]^2 + B[f'(x)]^2$$

$$g'(x) = A \cdot 2f(x) \cdot f'(x) + B \cdot 2f'(x) \cdot f''(x)$$

$$= 2f'(x)[Af(x) + Bf''(x)]$$

$$\text{令 } A=1, B=\frac{1}{2}, \text{ 有 } g'(x) = -x[f'(x)]^2$$

$$\therefore x < 0 \text{ 时 } g'(x) > 0, g(x) \uparrow; x > 0 \text{ 时 } g'(x) < 0, g(x) \downarrow$$

$$\Rightarrow g(x) = g(0)$$

$\therefore f(x)$ 和 $f'(x)$ 都有界

$$7. (1) f(x) = x + x^n \text{ 在 } (0, +\infty) \uparrow$$

$$f(0) = 0 < 1, f(1) = 2 > 1$$

$\therefore$ 恰有一正根

$$(2) x_n + x_n^n = 1$$

$$x_{n+1} + x_{n+1}^{n+1} = 1$$

若 $x_{n+1} \geq x_n$, 则 $x_{n+1}^{n+1} > x_n^n$, $x_{n+1} + x_{n+1}^{n+1} > x_n + x_n^n = 1$, 矛盾

$$\therefore x_{n+1} < x_n$$

$$\text{而 } x_n \in (0, 2)$$

$\therefore \{x_n\}$ 收敛

若 $\lim_{n \rightarrow \infty} x_n = x_0 < 1$, 则 $\lim_{n \rightarrow \infty} (x_n + x_n^n) = x_0 + \lim_{n \rightarrow \infty} x_0^n < 1 + 0 = 1$, 矛盾

若 $x_0 > 1$ 同理

$$\therefore \lim_{n \rightarrow \infty} x_n = x_0 = 1$$

Date / /

X-Talk

2012-2013 第一学期 第三次测试

$$\begin{aligned} 1. (1) &= \int x(x^7 - 7x^6 + 21x^5 - 35x^4 + 35x^3 - 21x^2 + 7x - 1) dx \\ &= \int (x^8 - 7x^7 + 21x^6 - 35x^5 + 35x^4 - 21x^3 + 7x^2 - x) dx \\ &= \frac{1}{9}x^9 - \frac{7}{8}x^8 + 3x^7 - \frac{35}{6}x^6 + 7x^5 - \frac{21}{4}x^4 + \frac{7}{3}x^3 - \frac{1}{2}x^2 + C \end{aligned}$$

$$\begin{aligned} (2) &= \int 2 \sin x \cos^3 x dx \\ &= \int -2 \cos^3 x d \cos x \\ &= -\frac{1}{2} \cos^4 x + C \end{aligned}$$

$$\begin{aligned} (3) \text{ 令 } t^2 = x, t = \sqrt{x} \\ \text{原式} &= 2 \int t \sin t dt = 2 \int t d \cos t \\ &= -2(t \cos t - \int \cos t dt) \\ &= -2t \cos t + 2 \sin t + C \\ &= -2\sqrt{x} \cos \sqrt{x} + 2 \sin \sqrt{x} + C \end{aligned}$$

$$\begin{aligned} (4) \text{ 令 } x = \sinh t \\ \text{原式} &= \int \ln(\sinh t + \cosh t) d \sinh t \\ &= \int t d \sinh t \\ &= t \sinh t - \int \sinh t dt \\ &= t \sinh t - \cosh t + C \end{aligned}$$

$$\begin{aligned} 2. (1) &= -\int_0^1 x d \arcsin x + x \arcsin x \Big|_0^1 \\ &= -\int_0^{\frac{\pi}{2}} \sin t dt + \frac{\pi}{2} = -\cos t \Big|_0^{\frac{\pi}{2}} + \frac{\pi}{2} = 1 + \frac{\pi}{2} \end{aligned}$$

$$\begin{aligned} (2) \text{ 令 } x = \tan t \\ \text{原式} &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{t}{\tan^2 t} \cdot \sec^2 t \cdot dt = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{t}{\sin^2 t} dt = -\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} t d \cot t \\ &= -t \cot t \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot t dt = \frac{\pi}{4} + \ln \sin t \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} = \frac{\pi}{4} + \ln \sqrt{2} \end{aligned}$$

Memo NO. _____

Date / /

X-Talk

$$(3) = \int_0^1 \left(\frac{x}{x^2+1} + \frac{1}{x+1} \right) dx$$

$$= \int_0^1 \frac{dx^2}{2(x^2+1)} + \int_0^1 \frac{dx}{x+1}$$

$$= -\frac{1}{2} \ln(x^2+1) \Big|_0^1 + \ln(x+1) \Big|_0^1 = \frac{1}{2} \ln 2$$

$$(4) \text{ 设 } t = \sqrt{1-\cos x} \therefore f(x) \text{ 关于 } x=\pi \text{ 是奇函数}$$

$$\therefore \text{原式} = 0$$

$$3. (1) \frac{1}{x^4} \cdot \frac{x^8}{1+x^6} = 0$$

$$(2) = \lim_{n \rightarrow \infty} \frac{1}{n^2} \cdot \frac{1}{n^2} = 0$$

$$\sum_{k=1}^n \frac{n}{n^2+k} < \sum_{k=1}^n \frac{n}{n^2} = 1$$

$$\sum_{k=1}^n \frac{n}{n^2+k} > \sum_{k=1}^n \frac{n}{n^2+n} = \frac{n}{n+1} \rightarrow 1 \quad (n \rightarrow \infty)$$

$$\therefore \text{原式} = 1$$

$$4. S = \int_0^{+\infty} \sqrt{r^2 + r'^2} d\theta$$

$$= \int_0^{+\infty} \sqrt{2} e^{-\theta} d\theta = \int_0^{+\infty} \sqrt{2} \cdot e^{-\theta} d\theta = -\sqrt{2} e^{-\theta} \Big|_0^{+\infty} = \sqrt{2}$$

$$5. I^2 = \left(\int_0^1 f(x) dx \right)^2 \leq \left(\int_0^1 x^2 dx \right) \left(\int_0^1 f^2(x) dx \right) = \frac{1}{3} \cdot \frac{1}{3} (6I-1)$$

$$9I^2 - 6I + 1 \leq 0 \Rightarrow "="$$

$$\therefore \frac{f(x)}{x} = C \quad \text{代入得 } C=1$$

$$\Rightarrow f(x) = x$$

$$6. g(x, y) = \int_0^x f(t+y) dt - \int_0^x f(t) dt$$

$$= \int_y^{x+y} f(t) dt - \int_0^x f(t) dt$$

$$= \left(\int_y^{x+y} f(t) dt + \int_x^y f(t) dt \right) - \left(\int_0^x f(t) dt + \int_x^y f(t) dt \right)$$

$$= \int_x^{x+y} f(t) dt - \int_0^y f(t) dt = g(y, x)$$

Memo NO. _____

Date / /

X-Talk

2013-2014 第一学期 第三次测试

$$1. (1) = \int \frac{(x-3)+8}{(x-3)^2+4} d(x-3)$$

$$\text{设 } x-3 = 2 \tan t$$

$$\text{原式} = \int (\tan t + 4) \cos^2 t dt$$

$$= \int (\tan t + 4) dt$$

$$= -\ln \cos t + 4t + C$$

$$(2) \text{ 设 } \sqrt{1-x} = t, x = 1-t^2$$

$$\text{原式} = \int_0^1 (1-t^2) t d(1-t^2)$$

$$= \int_1^0 2t^2(t^2-1) dt = \int_1^0 (2t^4 - 2t^2) dt$$

$$= \left(\frac{2}{5} t^5 - \frac{2}{3} t^3 \right) \Big|_1^0 = \frac{4}{15}$$

$$(3) = -\int_1^{+\infty} \ln x d \frac{1}{1+x} = -\frac{\ln x}{1+x} \Big|_1^{+\infty} + \int_1^{+\infty} \frac{1}{(1+x)x} dx$$

$$= \int_1^{+\infty} \left(\frac{1}{x} - \frac{1}{x+1} \right) dx = (\ln x - \ln(x+1)) \Big|_1^{+\infty} = \ln \frac{x}{x+1} \Big|_1^{+\infty} = \ln 2$$

$$(4) = \begin{cases} \frac{1}{3} x^3 - \frac{2}{3} + C, & x \leq -1 \\ x + C, & -1 < x < 1 \\ \frac{1}{4} x^4 + \frac{3}{4} + C, & x \geq 1 \end{cases}$$

$$2. (1) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n \sqrt{1+\frac{3k}{n}}} = \int_0^1 \frac{1}{\sqrt{1+3x}} dx$$

$$= \frac{2}{3} \sqrt{1+3x} \Big|_0^1 = \frac{2}{3}$$

$$(2) = \lim_{x \rightarrow +\infty} a \frac{(\ln \frac{x}{3})^n}{\frac{x}{3}+2} = \begin{cases} 0, & n=0, 1 \\ \lim_{x \rightarrow +\infty} a n \frac{(\ln \frac{x}{3})^{n-1}}{\frac{x}{3}} = 0, & n \geq 2 \end{cases} = 0, n \in \mathbb{N}$$

$$3. \begin{cases} x = 2y^2 \\ x = 1+y^2 \end{cases} \Rightarrow (x, y) = (2, \pm 1)$$

$$S = \int_{-1}^1 (4y^2 - 2y^2) dy = \int_{-1}^1 (1-y^2) dy = (y - \frac{1}{3} y^3) \Big|_{-1}^1 = \frac{4}{3}$$

Memo NO. _____

X-Talk

Date / /

$$4. (1) V_1 = \int_a^2 \pi y^2 dx = 4\pi \int_a^2 x^4 dx = 4\pi \cdot \frac{1}{5} x^5 \Big|_a^2 = \frac{4\pi}{5} (32 - a^5)$$

$$V_2 = \int_0^{2a^2} \pi (a^2 - x^2) dy = \int_0^{2a^2} \pi (a^2 - \frac{y}{2}) dy$$

$$= (\pi a^2 y - \frac{\pi y^2}{4}) \Big|_0^{2a^2} = 2\pi a^4 - \pi a^4 = \pi a^4$$

$$(2) V_1 + V_2 = \frac{4\pi}{5} (32 - a^5) + \pi a^4 \triangleq f(a)$$

$$f'(a) = -4\pi a^4 + (\frac{4\pi}{5}) a^3 = 4\pi a^3 (-a + 1)$$

$$\text{令 } f'(a) = 0 \Rightarrow a = 1$$

$$f(a) = f(1) = \frac{4\pi}{5} \times 31 + \pi = \frac{129\pi}{5}$$

5. ✓

$$\forall 6. \int_a^b x f(x) dx = \int_a^b x dF(x) = x F(x) \Big|_a^b - \int_a^b F(x) dx \quad (F(x) \triangleq \int_a^x f(t) dt)$$

$$= b \int_a^b f(t) dt - \int_a^b F(x) dx$$

$$\text{同理 } \int_a^b x g(x) dx = b \int_a^b g(t) dt - \int_a^b G(x) dx$$

$$\therefore \int_a^b x f(x) dx - \int_a^b x g(x) dx = \int_a^b [G(x) - F(x)] dx \leq 0$$

$$7. F(x) \triangleq \int_a^x f(x) dx \Rightarrow F(a) = F(b) = 0$$

$$\exists x_0 \in (a, b), \text{ s.t. } f(x_0) = F'(x_0) = 0$$

$$\mathcal{F}(x) \triangleq \int_a^x F(x) dx$$

$$0 = \int_a^b x f(x) dx = b \int_a^b f(t) dt - \int_a^b F(x) dx = -\mathcal{F}(b)$$

$$\mathcal{F}(a) = \mathcal{F}(b) = 0 \Rightarrow \exists x_0 \in (a, b), \text{ s.t. } F(x_0) = \mathcal{F}(x_0) = 0$$

$$\therefore \exists x_1 \in (a, x_0), x_2 \in (x_0, b), \text{ s.t. } f(x_1) = f(x_2) = 0$$

Memo NO. _____

X-Talk

Date / /

[2] 2015-2016 第一学期 第三次测试

$$1. (1) = \int (t+1) t^5 dt = \int (t^6 + t^5) dt$$

$$= \frac{1}{7} t^7 + \frac{1}{6} t^6 = \frac{1}{7} (x-1)^7 + \frac{1}{6} (x-1)^6 + C$$

$$(2) \text{ 设 } x = \sin^2 t$$

$$\text{原式} = \int \frac{2 \sin t \cos t}{\sin t \cos t} dt = 2t + C = 2 \arcsin \sqrt{x} + C$$

$$(3) = \int \frac{1}{2 \sin x (1 + \cos x)} dx = \int \frac{1}{8} \frac{(t^2+1)(t+\frac{1}{t})}{(t^2+1)^2} d(\arctan 2t) \quad (t = \tan \frac{x}{2})$$

$$= \int \frac{2(t^2+1)^2}{8(4t^2+1)t} dt$$

$$= \int \frac{\sin x dx}{2(1-\cos^2 x)(1+\cos x)} = \int \frac{du}{2(1-u^2)(1+u)} \quad (u = \cos x)$$

$$= \int \frac{1}{8} \left(\frac{u+3}{(u+1)^2} - \frac{1}{u-1} \right) du = \int \frac{1}{8} \left(\frac{1}{u+1} + \frac{2}{(u+1)^2} - \frac{1}{u-1} \right) du$$

$$= -\frac{1}{8} \left(\ln(u+1) - \frac{2}{u+1} - \ln(u-1) \right) + C$$

$$= -\frac{1}{8} \left(\ln \frac{\cos x + 1}{-\cos x + 1} - \frac{2}{\cos x + 1} \right) + C$$

$$(4) = \int \ln x d(x^2 - x) = -\int (x^2 - x) d \ln x + \ln x \cdot (x^2 - x)$$

$$= -\int (x-1) dx + \ln x \cdot (x^2 - x) = -\frac{1}{2} x^2 + x + \ln x \cdot (x^2 - x) + C$$

$$2. (1) = -\int_0^1 x^2 de^{-x} = -x^2 e^{-x} \Big|_0^1 + \int_0^1 2e^{-x} x dx$$

$$= -\frac{1}{e} + 2 \int_0^1 -x de^{-x} = -\frac{1}{e} - 2x e^{-x} \Big|_0^1 + 2 \int_0^1 e^{-x} dx$$

$$= -\frac{3}{e} + (-2e^{-x}) \Big|_0^1 = 2 - \frac{5}{e}$$

$$(2) = \int_0^1 \left(\frac{x+2}{x^2+1} - \frac{1}{x+1} \right) dx = \int_0^{\frac{\pi}{4}} (\tan t + 2) dt - \int_0^1 \frac{dx}{x+1} \quad (x = \tan t)$$

$$= [-\ln(\cos t) + 2t] \Big|_0^{\frac{\pi}{4}} - \ln(x+1) \Big|_0^1 = -\frac{1}{2} \ln 2 + \frac{\pi}{2}$$

Date / /

X-Talk

$$(3) = \int_{-3}^{-\sqrt{2}} x^2 dx + \int_{-\sqrt{2}}^{\sqrt{2}} 2 dx + \int_{\sqrt{2}}^2 x^2 dx$$

$$= \frac{1}{3} x^3 \Big|_{-3}^{-\sqrt{2}} + 4\sqrt{2} + \frac{1}{3} x^3 \Big|_{\sqrt{2}}^2 = \frac{35+8\sqrt{2}}{3}$$

$$(4) = \int_{-1}^1 \frac{x^2 e^x}{e^x+1} dx = \int_{-1}^1 x^2 \frac{1}{e^x+1} dx$$

$$= x^2 \Big|_{-1}^1 + \int_{-1}^1 \frac{1}{e^x+1} d[x^2(e^x+1)] = \int_{-1}^1 (2x + \frac{x^2 e^x}{e^x+1}) dx$$

$$= x^2 \Big|_{-1}^1 + I = 2 + I$$

$$\therefore I = 0$$

$$= \int_{-1}^0 \frac{x^2}{e^{-x}+1} dx + \int_0^1 \frac{x^2 e^x}{e^x+1} dx$$

$$= \int_0^1 \frac{x^2}{e^x+1} dx + \int_0^1 \frac{x^2 e^x}{e^x+1} dx = \int_0^1 x^2 dx = \frac{1}{3} x^3 \Big|_0^1 = \frac{1}{3}$$

$$3.(1) = \lim_{n \rightarrow \infty} \left(\prod_{k=1}^n \left(1 + \frac{k}{n} \right) \right)^{\frac{1}{n}} = \exp \left[\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \ln \left(1 + \frac{k}{n} \right) \right]$$

$$= \exp \left[\int_0^1 \ln(1+x) dx \right] = \exp \left[(1+x)(\ln(1+x)-1) \Big|_0^1 \right]$$

$$= e^{2\ln 2 - 1} = \frac{4}{e}$$

$$(2) = \lim_{x \rightarrow 0} \frac{x \int_0^x f(t) dt - \int_0^x t f(t) dt}{x \int_0^x f(t) dt}$$

$$= 1 - \lim_{x \rightarrow 0} \frac{\int_0^x t f(t) dt}{x \int_0^x f(t) dt} = 1 - \lim_{x \rightarrow 0} \frac{x f(x)}{\int_0^x f(t) dt + x f(x)}$$

$$= 1 - \frac{1}{\lim_{x \rightarrow 0} \frac{\int_0^x f(t) dt}{x f(x)} + 1} = 1 - \frac{1}{\lim_{x \rightarrow 0} \frac{\frac{f(x)}{f(x)}}{\frac{\int_0^x f(t) dt}{x} + 1}}$$

$$= 1 - \frac{1}{\frac{f(0)}{f(0)} + 1} = \frac{1}{2}$$

Memo NO. _____

Date / /

X-Talk

$$4. \text{ 设 } g(x) = f(x) + f(-x)$$

$$0 = \int_{-1}^1 f(x) g(x) dx = \int_{-1}^1 f(-x) g(-x) d(-x) = \int_{-1}^1 f(-x) g(x) dx$$

$$\therefore \int_{-1}^1 [f(x) + f(-x)] g(x) dx = 0$$

$$\Rightarrow \int_{-1}^1 [f(x) + f(-x)]^2 dx = 0$$

$$\Rightarrow f(x) + f(-x) \equiv 0 \Rightarrow f(x) \text{ 奇}$$

$$5. \forall x \in [a, b], \text{ RHS} = |f(\xi)| + \int_a^b |f'(x)| dx \quad (\exists \xi \in [a, b])$$

$$\geq |f(\xi)| + \int_{\xi}^x |f'(t)| dt$$

$$\geq |f(\xi) + \int_{\xi}^x f'(t) dt|$$

$$= |f(\xi) + f(x) - f(\xi)| = |f(x)|$$

Date / /

X-Talk

22 2016-2017 第一学期第三次测试

1. $\forall \varepsilon > 0, \exists \delta > 0$, 当划分 $\|T\| < \delta$ 时, 任取 $x_i \in [a_i, a_{i+1}]$, 均有 $\left| \sum_{i=1}^n (a_{i+1} - a_i) f(x_i) - I \right| < \varepsilon$, 则 $f(x)$ 在 $[a_1, a_{n+1}]$ 上 Riemann 可积.

2. (1) $t \triangleq 1 + \sqrt{x}$, $x = (t-1)^2$

$$\begin{aligned}\text{原式} &= \int t^{100} d(t-1)^2 = \int (2t^{101} - 2t^{100}) dt = \frac{1}{51} t^{102} - \frac{2}{101} t^{101} + C \\ &= \frac{1}{51} (1+\sqrt{x})^{102} - \frac{2}{101} (1+\sqrt{x})^{101} + C\end{aligned}$$

(2) $x \triangleq \sin t$

$$\text{原式} = \int \frac{t}{\cos t} \cdot \frac{1 + \sin^2 t}{\sin^2 t} \cdot \cos t \cdot dt = \int \frac{t(1 + \sin^2 t)}{\sin^2 t} dt$$

$$= \int (t \csc^2 t + t) dt = -\int t d \cot t + \int t dt$$

$$= -t \cot t + \int \cot t \, dt + \frac{1}{2} t^2$$

$$= -t \cot t + \ln(\sin t) + \frac{1}{2} t^2 + C$$

$$= -\arcsin x \cdot \frac{\sqrt{1-x^2}}{x} + \ln x + \frac{1}{2} \arcsin^2 x + C$$

$$3.(1) = \frac{1}{2} \left(\int_0^{n\pi} x |\sin x| dx + \int_0^{n\pi} (n\pi - x) |\sin x| dx \right) = \frac{1}{2} \int_0^{n\pi} n\pi |\sin x| dx$$

$$= \frac{1}{2} n^2 \pi \int_0^\pi \sin x \, dx = -\frac{1}{2} n^2 \pi \cos x \Big|_0^\pi = n^2 \pi$$

~~$$(2) = \frac{1}{2} \left(\int_0^1 \frac{x}{e^x + e^{\frac{1}{2}-x}} dx + \int_0^1 \frac{1-x}{e^{1-x} + e^{1+x}} dx \right)$$~~

$$= \int_0^1 \frac{2}{e^x + e^{2-x}} dx = \int_1^e \frac{2}{y + e^2/y} d \ln y + \int_1^e \frac{2}{y^2 + e^2} dy = \int_{\frac{1}{e}}^1 \frac{2}{e(t^2+1)} dt = \frac{2}{e} \arctan t \Big|_{\frac{1}{e}}^1 = \frac{2}{e} \left(\frac{\pi}{4} - \arctan \frac{1}{e} \right)$$

$$(3) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos t \, dt}{(2 - \sin^2 t) \cos t} = 2 \int_0^{\frac{\pi}{2}} \frac{\sin^2 t + \cos^2 t}{\sin^2 t + 2 \cos^2 t} dt = 2 \int_0^{\frac{\pi}{2}} \frac{p^2 + 1}{p^2 + 2} \cdot \frac{dp}{p^2 + 1}$$

$$(x = \sin t, p = \tan t) = 2 \int_0^1 \frac{1}{p^2 + 2} dp = \sqrt{2} \arctan \frac{p}{\sqrt{2}} \Big|_0^1$$

$$= \sqrt{2} \arctan \frac{\sqrt{2}}{2}$$

Memo NO. _____

Date / /

X-Talk

证4. 不妨设单调递增. 当 $\frac{k}{n} \leq x \leq \frac{k+1}{n}$ 时有 $f(\frac{k}{n}) \leq f(x) \leq f(\frac{k+1}{n})$ 累加得...

由图知 $\sum_{k=1}^{n-1} f(\frac{k}{n}) \cdot \frac{1}{n} \leq \int_0^1 f(x) dx \leq \sum_{k=1}^n f(\frac{k}{n}) \cdot \frac{1}{n}$

$$\text{令 } n \rightarrow \infty, \text{ 有 } \lim_{n \rightarrow \infty} \sum_{k=1}^n f\left(\frac{k}{n}\right) \cdot \frac{1}{n} = \lim_{n \rightarrow \infty} \sum_{k=1}^{n-1} f\left(\frac{k}{n}\right) \cdot \frac{1}{n}$$

$$\leq \int_0^1 f(x) dx \leq \lim_{n \rightarrow \infty} \sum_{k=1}^n f\left(\frac{k}{n}\right) \cdot \frac{1}{n} \quad \therefore \text{取等号}$$

5. $F'(0) = \lim_{t \rightarrow 0} t \sin \frac{1}{t} = 0$

$$G'(0) = \lim_{x \rightarrow 0} \frac{\int_0^x \cos \frac{1}{t} dt}{x} = \lim_{x \rightarrow 0} \frac{\int_{\frac{1}{x}}^{\infty} \frac{1}{p^2} \cos p dp}{x}$$

$$= \lim_{x \rightarrow 0} \frac{-\int_0^x t^2 d \sin \frac{1}{t}}{x} = \lim_{x \rightarrow 0} \frac{-x^2 \sin \frac{1}{x} + 2 \int_0^x t \sin \frac{1}{t} dt}{x} = \lim_{x \rightarrow 0} \frac{2 \int_0^x t^3 d \cos \frac{1}{t}}{x} = \lim_{x \rightarrow 0} \frac{2x^3 \cos \frac{1}{x} - 6 \int_0^x t^2 \cos \frac{1}{t} dt}{x}$$

$$= \lim_{x \rightarrow 0} \frac{-6x^2 \cos \frac{1}{x}}{x} = 0$$

6. (1) 不一定. $f(x) = \begin{cases} 0, & a \leq x \leq \frac{a+b}{2} \\ 1, & \frac{a+b}{2} < x \leq b \end{cases}$

(2) 不一定. $f(x) = x \sin \frac{1}{x}, x \in [0, 1]$

$$7. I_n = \int_0^\pi \cos^n x \cdot \cos nx dx = \int_0^\pi \cos^{n-1} x \cos nx d \sin x$$

~~$$= -\int_0^{\pi} \sin x d(\cos^{n-1} x \cdot \cos nx) = -\int_0^{\pi} \sin^2 x \cdot (n-1) \cos^{n-2} x \cos nx$$~~

$$= \int_0^\pi \left(\frac{e^{ix} + e^{-ix}}{2} \right)^n \cdot \frac{e^{inx} + e^{-inx}}{2} dx = \int_0^\pi \frac{1}{2^{n+1}} \left(\sum_{k=0}^{2n} C_{2n}^k e^{i(k-n)x} \right) \cdot (e^{inx} + e^{-inx}) dx$$

$$= \int_0^\pi \frac{1}{2^{n+1}} \sum_{k=0}^{2n} C_{2n}^k (e^{ikx} + e^{-ikx}) dx = \frac{1}{2^{n+1}} \left[\sum_{k=1}^{2n} \left(\frac{1}{ik} e^{ikx} - \frac{1}{ik} e^{-ikx} \right) + \frac{1}{2^n} \sum_{k=0}^{2n} C_{2n}^k \sin kx \right] \Big|_0^\pi = \frac{\pi}{2^n}$$

$$8. \text{LHS} = \int_a^1 (e^{-x} f(x))' e^x dx = \int_a^1 (e^{-x} f(x))' dx$$

$$= \int_0^1 e^x d(e^{-x} f(x)) = f(x) \Big|_0^1 - \int_0^1 e^{-x} f(x) dx$$

$$\downarrow = f(1) - f(0) - \int_0^1 f(x) dx \quad \times$$

$$\text{LHS} \rightarrow \int_0^1 (e^x f(x))' dx = e^x f(x) \Big|_0^1 = \frac{1}{e}$$

缩放

(注: $F(x) = e^{-x} f(x)$
 $F(0) = 0, F(1) = \frac{1}{e}$)

$$\exists \xi, f'(\xi) > 0$$

不变号 $\Rightarrow f'(x) > 0$

Memo NO. _____

X-Talk

Date / /

19 2012-2013 第一学期第四次测试

1. (1) $\frac{dy}{dx} = \frac{x+2}{y-2}$

$$(y-2)dy = (x+2)dx$$

$$\frac{1}{2}y^2 - 2y = \frac{1}{2}x^2 + 2x + C$$

(2) $\begin{cases} x+y+2=0 \\ x-y+4=0 \end{cases} \Rightarrow \begin{cases} x_0 = -3 \\ y_0 = 1 \end{cases}$

$$\begin{cases} u = x+3 \\ v = y-1 \end{cases} \Rightarrow \frac{dv}{du} = \frac{u+v}{u-v}$$

$$\text{令 } k = \frac{v}{u}, v = ku, \frac{dv}{du} = k + \frac{dk}{du} \Rightarrow k + \frac{dk}{du} = \frac{1+k}{1-k}$$

$$\frac{1-k}{1+k^2} dk = du$$

$$\therefore u = \arctan k - \frac{1}{2} \ln(1+k^2) + C$$

$$= \arctan \frac{v}{u} - \frac{1}{2} \ln(1 + \frac{v^2}{u^2}) + C$$

$$\text{即 } x = \arctan \frac{y-1}{x+3} - \frac{1}{2} \ln(1 + \frac{(y-1)^2}{(x+3)^2}) + C'$$

2. 令 $z = y'$, $z' = y'' \Rightarrow z' - xz - x^3 z^3 = 0$

$$z^{-3} z' - xz^{-2} = x^3$$

$$\text{令 } u = z^{-2} \Rightarrow \frac{du}{dx} + 2xu = -2x^3 \quad (*)$$

$$\frac{du}{dx} = -2xu \Rightarrow \frac{du}{u} = -2x dx \Rightarrow \ln u = -x^2 + C$$

$$y'(0) = 1 \Rightarrow u = e^{-x^2}$$

将 $u = ve^{-x^2}$ 代入 (*), 得 $\frac{dv}{dx} e^{-x^2} = -2x^3$

$$dv = -x^2 e^{x^2} dx^2 = e^t t dt \quad (t = x^2)$$

$$v = -\int t de^t = -te^t + \int e^t dt = (1-t)e^t + C$$

$$\therefore u = 1 - x^2 + Ce^{-x^2}$$

$$y'(0) = 1 \Rightarrow u = 1 - x^2 \Rightarrow z = \pm \sqrt{1-x^2}$$

Memo NO. _____

Date / /

X-Talk

$$\frac{dy}{dx} = \pm \frac{1}{\sqrt{1-x^2}}$$

$$y = \pm \int \frac{1}{\sqrt{1-x^2}} dx = \pm \int \frac{1}{\sqrt{1-\sin^2 t}} d\sin t = \pm \int dt = \pm \arcsin x + C$$

$$y(0) = 0 \Rightarrow y = \pm \arcsin x$$

3. (1) $\sim \sum_{n=1}^{\infty} \frac{\sqrt{2n\pi}}{e^n}$ 收敛

(2) $\int_2^{+\infty} \frac{1}{x(\ln x)^2} dx = -\frac{1}{\ln x} \Big|_2^{+\infty} = \frac{1}{\ln 2}$ 收敛

(3) $< \sum_{n=1}^{\infty} \frac{n}{2^n}$ 收敛

(4) $= \sum_{n=1}^{\infty} \frac{(-1)^n}{[\ln(2n+1)]^3}$ 条件收敛

(5) $\frac{1}{\sqrt{1+x}} = 1 + \sum_{n=1}^{\infty} (-1)^n \frac{(2n-1)!!}{(2n)!!} x^n$

$$a_n > \frac{(2n-2)!!}{(2n)!!} = \frac{1}{2n}$$

$$\text{令 } x = 1 \text{ 得 } = \frac{\sqrt{2}}{2} - 1$$

$$\frac{a_{n+1}}{a_n} = \frac{2n+1}{2n+2} < 1 \text{ 而 } \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{1}{2} > 0 \therefore \text{条件收敛}$$

4. (1) $\lim_{n \rightarrow \infty} f_n(0) = 0$

$$\text{当 } 0 < x \leq 1, \lim_{n \rightarrow \infty} f_n(x) = 0$$

$$\therefore \text{收敛 } \alpha \in \mathbb{R}$$

(2) $f'_n(x) = n^\alpha (2xe^{-nx} - nx^2 e^{-nx}) = n^\alpha x e^{-nx} (2-nx)$ 用导数找最值

$$\lim_{n \rightarrow \infty} \sup_{x \in [0,1]} f_n(x) = \lim_{n \rightarrow \infty} f_n(\frac{2}{n}) = \lim_{n \rightarrow \infty} n^{\alpha-2} \cdot \frac{4}{e^2} = 0$$

$$\therefore \alpha < 2$$

(3) RHS = 0

$$\text{LHS} = 0 \Leftrightarrow \forall \varepsilon > 0, \exists N > 0, \forall n > N, \forall x \in [0,1], |f_n(x)| < \varepsilon \Leftrightarrow |f_n(x)| < \varepsilon, \forall x$$

$$\Leftrightarrow \alpha < 2$$

直接积

$$\int_0^1 f_n(x) dx < \varepsilon \Leftrightarrow \int_0^1 |f_n(x)| dx < \varepsilon$$

Date / /

$$5. \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{n+2}{n+3} = 1$$

$$\therefore R=1$$

当 $x=1$ 时不收敛, 当 $x=-1$ 时条件收敛

$$\therefore -1 \leq x < 1$$

$$\therefore \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \Rightarrow \frac{x^2}{1-x} = \sum_{n=0}^{\infty} x^{n+2}$$

$$\therefore \text{求导, 得 } \frac{2x-x^2}{(1-x)^2} = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+2} \Rightarrow \frac{2-x}{(1-x)^2} = \sum_{n=0}^{\infty} \frac{x^n}{n+2}$$

$$6. y = \arctan \frac{3x-1}{3x+1}$$

$$\tan y = \frac{3x-1}{3x+1} \Rightarrow y = \arctan 3x - \frac{\pi}{4}$$

$$\text{由 } \arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} \quad (-1 \leq x \leq 1) \quad (*)$$

$$\text{得 } y = \sum_{n=0}^{\infty} \frac{(-1)^n (3x)^{2n+1}}{2n+1} - \frac{\pi}{4} \quad \left(-\frac{1}{3} < x \leq \frac{1}{3}\right)$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \arctan 1 = \frac{\pi}{4}$$

$$4(3) \int_0^1 n^\alpha x^2 e^{-nx} dx = \frac{n^\alpha}{n} \int_0^1 x^2 d e^{-nx}$$

$$= -n^{\alpha-1} (x^2 e^{-nx} \Big|_0^1 - \int_0^1 e^{-nx} dx)$$

$$= -n^{\alpha-1} (e^{-n} - 2 \int_0^1 x e^{-nx} dx)$$

$$= -n^{\alpha-1} (e^{-n} + \frac{2}{n} \int_0^1 x d e^{-nx})$$

$$= -n^{\alpha-1} (e^{-n} + \frac{2}{n} x e^{-nx} \Big|_0^1 - \frac{2}{n} \int_0^1 e^{-nx} dx)$$

$$= -n^{\alpha-1} \left[\left(\frac{2}{n} + 1 \right) e^{-n} + \frac{2}{n^2} e^{-nx} \Big|_0^1 \right]$$

$$= -n^{\alpha-1} \left[\left(\frac{2}{n} + 1 \right) e^{-n} + \frac{2}{n^2} (e^{-n} - 1) \right] \rightarrow \frac{2}{n^{3-\alpha}}$$

$$\therefore \alpha < 3$$

Date / /

29 2011-2012 第一学期期末考试

$$1. (1) = \lim_{x \rightarrow 0} \frac{\frac{1}{2} x^2}{x^2} = \frac{1}{2}$$

$$(2) = \lim_{x \rightarrow +\infty} \left(\left(1 + \frac{3}{x-1} \right)^{\frac{x-1}{3}} \right)^3 \left(1 + \frac{3}{x-1} \right) = e^3$$

$$(3) = \lim_{n \rightarrow \infty} \frac{\sin^2 \xi}{\xi} \quad (\xi \in (n, n+1))$$

$$= \lim_{\xi \rightarrow \infty} \frac{\sin^2 \xi}{\xi} = 0$$

$$(4) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{1}{1 + \left(\frac{k}{n} \right)^2} = \int_0^1 \frac{1}{1+x^2} dx = \arctan x \Big|_0^1 = \frac{\pi}{4}$$

$$2. (1) = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{\sqrt{\sin^2 t (1 - \sin^2 t)}} d \sin^2 t = 2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} dt = \frac{\pi}{2}$$

$$(2) = \int_0^{+\infty} \frac{1}{2} \left(\frac{1}{1+x} + \frac{1-x}{1+x^2} \right) dx$$

$$= \frac{1}{2} \left(\int_0^{+\infty} \frac{d(1+x)}{1+x} + \int_0^{+\infty} \frac{dx}{1+x^2} - \frac{1}{2} \int_0^{+\infty} \frac{d(1+x^2)}{1+x^2} \right)$$

$$= \frac{1}{2} \left[\ln(1+x) + \arctan x - \frac{1}{2} \ln(1+x^2) \right] \Big|_0^{+\infty}$$

$$= \frac{1}{2} \arctan x \Big|_0^{+\infty} = \frac{\pi}{4}$$

$$3. (1) a_n = \frac{n+1}{3n-2}$$

$$\lim_{n \rightarrow \infty} a_n = \frac{1}{3} \neq 0$$

$\therefore$ 不收敛 (即发散)

$$(2) a_n = \frac{n^{n+1}}{n! e^n} = \frac{n^{n+1}}{\sqrt{2\pi n} \left(\frac{n}{e} \right)^n e^n e^{\theta_n/12n}} = \frac{1}{n \sqrt{2\pi n} e^{\theta_n/12n}}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{n^{3/2}} = \frac{1}{\sqrt{2\pi}} \quad \text{而 } \sum_{n=1}^{\infty} \frac{1}{n^{3/2}} \text{ 收敛}$$

$\therefore$ 收敛

Date _____
单调有界 $\Rightarrow$ 收敛

4. 若 $\lim_{x \rightarrow -\infty} f'(x) = a > 0$

则 $\forall \varepsilon > 0, \exists x_0 < 0, \forall x < x_0, |f'(x) - a| < \varepsilon$

即 $\varepsilon = a$, 则 $f'(x) > a, \forall x < x_0$

由 $f''(x) > 0$ 知 $f'(x) > a, \forall x \in \mathbb{R}$

$$f(x) = f(0) + \int_0^x f'(t) dt < f(0) - \int_x^0 a dt \quad (\text{当 } x < 0 \text{ 时})$$

$$= f(0) + ax$$

取 $x_1 = -\frac{f(0)}{a}$, 有 $f(x_1) < f(0) + ax_1 = 0$, 矛盾

$\therefore \lim_{x \rightarrow -\infty} f'(x) = 0$

5. 设 $z = y'$, $z' = y''$, 则 $\cancel{z' + z} z' - z = (2x+2)e^x$ (*)

$$z' - z = 0 \Rightarrow z = Ce^x$$

将 $z = ue^x$ 代入 (*) 得 $u'(x)e^x = (2x+2)e^x \Rightarrow u(x) = x^2 + 2x$

$$\therefore y' = (x^2 + 2x)e^x$$

$$y = \int (x^2 + 2x)e^x dx = \int (x^2 + 2x) de^x$$

$$= (x^2 + 2x)e^x - \int e^x (2x + 2) dx$$

$$= (x^2 + 2x)e^x - (2x + 2)e^x + \int 2e^x dx$$

$$= x^2 e^x + C$$

$$6. (1) \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} e^{(2n+1)x} < 1 \Rightarrow x > 0$$

$x = 0$ 时发散

$\therefore x > 0$

(2) 否, 取 $x = \frac{1}{2n+1}$, 则 $\lim_{n \rightarrow \infty} a_n = \frac{1}{e} \neq 0$

故 (2) 是. $\{\frac{1}{n}\}$ 单调递减, 一致趋于 0.

$$\sum_{k=1}^n \frac{1}{e^{kx}} < \sum_{k=1}^n \frac{1}{e^{kx}} = \frac{1}{e^x} \sum_{k=1}^n \left(\frac{1}{e}\right)^k < \frac{1}{e^x} \cdot \frac{1}{1 - \frac{1}{e}} < \frac{1}{e-1}$$

$$\therefore \sum_{k=1}^n \frac{1}{e^{kx}} \text{ 一致有界}$$

由 Dirichlet 判别法知其一致收敛.

(3) 由一致收敛级数的性质可知.

7. 由 Cauchy 准则, 有 $\forall \varepsilon > 0, \exists N > 0, \forall n > N, p > 0$, 有

$$\sum_{i=n+1}^{n+p} \left(\frac{a_{i+1}}{a_i} - 1 \right) = \sum_{i=n+1}^{n+p} \frac{a_{i+1} - a_i}{a_i} < \varepsilon$$

$$\text{LHS} \geq \frac{1}{a_{n+p+1}} \sum_{i=n+1}^{n+p} (a_{i+1} - a_i) = \frac{a_{n+p+1} - a_{n+1}}{a_{n+p+1}} = 1 - \frac{a_{n+1}}{a_{n+p+1}}$$

取 $\varepsilon = \frac{1}{2}$ 得 $a_{n+p+1} \leq 2a_{n+1}$

选定 n , 变动 p , 可知 $\{a_n\}$ 有界.

$$8. \text{LHS} = \int_0^1 |(e^x f(x))' e^{-x}| dx \geq \frac{1}{e} \int_0^1 d(e^x f(x))$$

$$= \frac{1}{e} (e^x f(x)) \Big|_0^1 = 1$$

30 2012-2013 第一学期期末考试

1. ~~$\exists \alpha \in \mathbb{R}, \forall \varepsilon > 0, \exists X_0 > 0, \forall x > X_0, \text{有 } |f(x) - \alpha| < \varepsilon$~~

$\forall \varepsilon > 0, \exists X_0 > 0, \forall x_1, x_2 > X_0, \text{有 } |f(x_1) - f(x_2)| < \varepsilon$

$$2. (1) \lim_{x \rightarrow +\infty} \frac{1}{4x^3} \cdot \frac{x^3 + |\cos x^2|}{1 + x^6} \cdot 2x = \lim_{x \rightarrow +\infty} \frac{x^3}{2x^3(1+x^6)} = \frac{1}{2}$$

(2) ✓

$$3. (1) = \int_0^{\frac{\pi}{2}} \frac{\sin 2x dx}{3 + \cos 2x} = \int_0^{\frac{\pi}{2}} \frac{d \cos 2x}{3 + \cos 2x} = \ln(3 + \cos 2x) \Big|_0^{\frac{\pi}{2}} = -\ln 2$$

$$(2) I_n = \int_0^{+\infty} x^{2n+1} e^{-x^2} dx = -\frac{1}{2} \int_0^{+\infty} x^{2n} d e^{-x^2}$$

$$= -\frac{1}{2} x^{2n} e^{-x^2} \Big|_0^{+\infty} + \frac{1}{2} \int_0^{+\infty} e^{-x^2} d x^{2n}$$

$$= n \int_0^{+\infty} x^{2n-1} e^{-x^2} dx = n I_{n-1} = n! I_0$$

$$= n! \int_0^{+\infty} \left(\frac{1}{2} e^{-x^2}\right) d(-x^2) = -\frac{1}{2} n! e^{-x^2} \Big|_0^{+\infty} = \frac{1}{2} n!$$

$$4. (1) \frac{dy}{dx} + 1 + \frac{y}{x} = 0$$

$$\text{令 } u = \frac{y}{x}, \frac{dy}{dx} = \frac{du}{dx} x + u \Rightarrow \frac{du}{dx} x + 2u + 1 = 0$$

$$\therefore \frac{du}{2u+1} = -\frac{dx}{x}$$

$$\text{积得 } \frac{1}{2} \ln(2u+1) = -\ln x + \ln C'$$

$$\text{即 } \left(\frac{2y}{x} + 1\right)^2 x = C'$$

$$y = \frac{1}{2} (\pm \sqrt{C'x} - x)$$

$$2. (2) z \triangleq y' \Rightarrow z' + z^2 = z$$

$$y'' = y' \frac{dy'}{dy}$$

$$y' e^y = e^y + C$$

$$\frac{dz}{z-z^2} = dx$$

$$y' \frac{dy'}{dy} + (y')^2 = y'$$

$$y = x+1$$

$$x = \int \frac{dz}{z(1-z)} = \int \frac{1}{z} dz - \int \frac{dz}{z-1} = \ln|z| - \ln|z-1| + C$$

$$\frac{z}{z-1} = C' e^x \text{ 即 } 1 - \frac{1}{z} = C'' e^{-x} \quad z(0)=1 \Rightarrow C''=0$$

$$5. (1) \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1 \Rightarrow R=1$$

$x=1$ 时收敛; $x=-1$ 时发散

$\therefore$ 收敛域 $(-1, 1]$

$$(2) \text{ 设该级数为 } f(x), \text{ 则 } f(x) = x(x+1) \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{n-1}}{n} = x(x+1) g(x)$$

$$\therefore \sum_{n=1}^{\infty} (-x)^n = x \cdot \frac{1}{1-(-x)} = \frac{x}{1+x}$$

$$\therefore g(x) = \left(\frac{x}{1+x} \right)' = \frac{1}{(1+x)^2}$$

$$\therefore f(x) = \frac{x}{1+x}$$

$$(3) |u_n| = \left| \frac{(x+1)x^n}{n} \right| \leq \frac{2}{n}$$

$\{(x+1)x^n\}$ 单调 $\downarrow$, 且 $|(x+1)x^n| \leq 2$ (一致有界)

$$\times \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \text{ 一致收敛}$$

$\therefore$ 级数一致收敛

$$6. g(x) = e^{cx} f(x)$$

$$g(a) = g(b) = 0$$

$$\text{微分中值定理} \Rightarrow g'(\xi) = e^{c\xi} (f'(\xi) + cf(\xi)) = 0 \Rightarrow \checkmark$$

$$7. (1) \Leftarrow (\beta - \alpha) \int_{\alpha}^{\beta} f(x) dx + (\beta - \alpha) \int_{\beta}^1 f(x) dx \geq (1 - \alpha) \int_{\alpha}^{\beta} f(x) dx$$

$$\Leftarrow \frac{1}{1-\beta} \int_{\beta}^1 f(x) dx \geq \frac{1}{\beta-\alpha} \int_{\alpha}^{\beta} f(x) dx$$

$$\Leftarrow f(\xi_2) \geq f(\xi_1) \quad (\xi_1 \in (\alpha, \beta), \xi_2 \in (\beta, 1))$$

$$\Leftarrow \xi_2 \geq \xi_1$$

$$? (2) \text{ 令 } \alpha \rightarrow 0, \beta \rightarrow 1, \text{ 则 } \int_{\alpha}^{\beta} f(x) dx \rightarrow \int_0^1 f(x) dx, \frac{1-\alpha}{\beta-\alpha} \rightarrow 1 \text{ 不能更大}$$

Memo NO. _____

Date / /

X-Talk

31 2013-2014 第一学期期末考试

$$1. (1) = \frac{1}{1+e^{x \sin x}} \cdot e^{x \sin x} \cdot (\sin x + x \cos x)$$

$$(2) = (x^2+1) \sin^{(n)} x + n \cdot 2x \cdot \sin^{(n-1)} x + n(n-1)x \cdot \sin^{(n-2)} x = \begin{cases} \text{(按 mod 4 分类讨论)} \end{cases}$$

$$2. (1) \lim_{n \rightarrow \infty} n \cdot \frac{\arctan \frac{1}{n}}{\frac{1}{n}} \quad (\xi \in (n^2, n^2+n))$$

$$= \lim_{n \rightarrow \infty} n \cdot \frac{\pi/2}{n^2} = 0$$

$$(2) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sqrt{1-(\frac{k}{n})^2}} \cdot \frac{1}{n} = \int_0^1 \frac{dx}{\sqrt{1-x^2}}$$

$$= \int_0^1 \frac{d \sin t}{\cos t} = \int_0^{\frac{\pi}{2}} dt = \frac{\pi}{2}$$

$$3. (1) = \int \ln x dx e^x = x e^x \ln x - \int x e^x d \ln x$$

$$= x e^x \ln x - \int e^x dx = x e^x \ln x - e^x + C$$

$$(2) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \cos x d \sin x = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin x \cos x)^2 dx$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\frac{1}{2} \sin 2x)^2 dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{8} (1 - \cos 4x) dx$$

$$= (\frac{1}{8} x - \frac{1}{32} \sin 4x) \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{\pi}{8}$$

$$4. \frac{y \cos x - y' \sin x}{y^2} = (1 - \sin x) \cos x$$

$$u' = \left(\frac{\sin x}{y} \right)' = (1 - \sin x) \cos x$$

$$u = \int du = \int (1 - \sin x) d \sin x = \sin x - \frac{1}{2} \sin^2 x + C$$

$$y = \frac{\sin x}{\sin x - \frac{1}{2} \sin^2 x + C}$$

$$5. \frac{x e^{-nx}}{\sqrt{n}} \leq \frac{x}{\sqrt{n}} \cdot \frac{1}{nx} = \frac{1}{n^{3/2}} \quad \text{而} \sum_{n=1}^{\infty} \frac{1}{n^{3/2}} \text{收敛}$$

$\therefore$ 一致收敛

Memo NO. _____

Date / /

$$7. \sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad (-1 < x < 1)$$

$$\sum_{n=0}^{\infty} x^{n+1} = \frac{x}{1-x} = \frac{1}{1-x} \quad (x \neq 0)$$

$$(4) \sum_{n=0}^{\infty} \frac{x^n}{n} = -\ln(1-x)$$

$$(5) \sum_{n=0}^{\infty} \frac{x^{n+1}}{n(n+1)} = \frac{x(\ln|x|+1)}{x} + (1-x)(\ln(1-x)-1) + 1$$

$$\therefore \sum_{n=1}^{\infty} \frac{x^n}{n(n+1)} = \frac{(1-x)(\ln(1-x)-1)+1}{x} \quad (-1 < x < 1 \text{ 且 } x \neq 0)$$

$x=0$ 时, 级数 $=0$; $x=1$ 时, 级数 $= \sum_{n=1}^{\infty} (\frac{1}{n} - \frac{1}{n+1}) = 1$; $x=-1$ 时, 上式仍成立.

收敛区域: $[-1, 1]$

$$7. f(x) = e^x + e^y + xy + (e^{xy} - 1)$$

$$f'(x) = e^x + y + y e^{xy}$$

$$f''(x) = e^x + y^2 e^{xy}$$

$$(e^x - 1)(e^y - 1) > (1+x-1)(1+y-1) = xy$$

8. (1) $\checkmark$

$$(2) \Leftarrow e^{dx} f'(x) \leq c e^{dx} f(x)$$

$$\Leftarrow (e^{dx} f'(x))' \leq c (e^{dx} f(x))'$$

$$\Leftarrow d f'(x) + f''(x) \leq c d f(x) + c f'(x)$$

$$\Leftarrow f''(x) \leq c d f(x) + (c-d) f'(x)$$

取 $cd=a$, $c-d=b$ 即可

32 2015-2016 第一学期期末考试

$$1. (1) = \lim_{n \rightarrow \infty} \sqrt{n} \cos n\xi \sin \xi^n = \lim_{n \rightarrow \infty} \sqrt{n} \xi^n \cos n\xi = 0 \quad (\xi \in (0, 1))$$

$$(2) = \lim_{x \rightarrow 0} \frac{\sin x^2 \cdot 2x}{\frac{4x^3}{1+x^4}} = \lim_{x \rightarrow 0} \frac{2x^3}{4x^3} = \frac{1}{2}$$

$$2. (1) = \frac{1}{3} \int \arctan x \, dx^3 = \frac{1}{3} x^3 \arctan x - \frac{1}{3} \int x^3 d \arctan x$$

$$= \frac{1}{3} x^3 \arctan x - \frac{1}{6} \int \frac{x^2}{1+x^2} dx^2$$

$$= \frac{1}{3} x^3 \arctan x - \frac{1}{6} \int dx^2 + \frac{1}{6} \int \frac{1}{1+x^2} d(1+x^2)$$

$$= \frac{1}{3} x^3 \arctan x - \frac{1}{6} x^2 + \frac{1}{6} \ln(1+x^2) + C$$

$$(2) = \int \frac{1}{\frac{1}{t}(1+\frac{1}{t^4})} d\frac{1}{t} = \int -\frac{t^3}{t^4+1} dt = -\frac{1}{4} \int \frac{d(t^4+1)}{t^4+1} = -\frac{1}{4} \ln(t^4+1) + C$$

$$= -\frac{1}{4} \ln\left(\frac{1}{x^4}+1\right) + C$$

$$(3) I = \int_0^1 \frac{(1-x)^2 e^x}{(1+x^2)^2} dx, J = \int_0^1 \frac{(1+x)^2 e^x}{(1+x^2)^2} dx$$

$$I+J = \int_0^1 \frac{2e^x}{1+x^2} dx, J-I = \int_0^1 \frac{4xe^x}{(1+x^2)^2} dx = -2 \int_0^1 e^x d \frac{1}{1+x^2} = -\frac{2e^x}{1+x^2} \Big|_0^1 + 2 \int_0^1 \frac{e^x}{1+x^2} dx = 2-e+I+J$$

$$\therefore I = \frac{1}{2}e - 1$$

$$(4) = \int_0^{+\infty} \left(\frac{1}{x+1} + \frac{-x+2}{x^2-x+1} \right) dx = \frac{1}{3} \left[\ln(x+1) \Big|_0^{+\infty} - \int_0^{+\infty} \frac{(x-\frac{1}{2}) - \frac{3}{2}}{(x-\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} dx \right]$$

$$\text{设 } x - \frac{1}{2} = \frac{\sqrt{3}}{2} y, \text{ 则 } \int_0^{+\infty} \frac{y - \frac{\sqrt{3}}{2}}{y^2 + 1} d\left(\frac{\sqrt{3}}{2} y + \frac{1}{2}\right)$$

$$= \frac{\sqrt{3}}{2} \int_{-\frac{1}{\sqrt{3}}}^{+\infty} \frac{y - \frac{\sqrt{3}}{2}}{y^2 + 1} dy = \frac{\sqrt{3}}{4} \int_{-\frac{1}{\sqrt{3}}}^{+\infty} \frac{dy^2}{y^2 + 1} - \frac{3}{2} \int_{-\frac{1}{\sqrt{3}}}^{+\infty} \frac{dy}{y^2 + 1}$$

$$= \left[\frac{\sqrt{3}}{4} \ln(y^2 + 1) - \frac{3}{2} \arctan y \right] \Big|_{-\frac{1}{\sqrt{3}}}^{+\infty} = \frac{\sqrt{3}}{4} \ln(y^2 + 1) \Big|_{-\frac{1}{\sqrt{3}}}^{+\infty} - \frac{3}{2} \left(\frac{\pi}{2} + \frac{\pi}{6} \right)$$

$$= \int \left(\frac{1}{1+x} + \frac{2-x}{x^2-x+1} \right) dx = \ln|1+x| - \frac{1}{2} \int \frac{d(x^2-x+1)}{(x^2-x+1)} + \frac{3}{2} \int \frac{dx}{x^2-x+1}$$

$$3. = x [\ln(1+x^2)]^{(n)} + n [\ln(1+x^2)]^{(n-1)} = n [\ln(1+x^2)]^{(n-1)}$$

$$\text{设 } g(x) = \ln(1+x^2)$$

$$g'(x) = \frac{2x}{1+x^2} \text{ 即 } (1+x^2) g'(x) = 2x$$

$$\therefore (1+x^2) g^{(k+1)}(x) + 2kx g^{(k)}(x) + k(k-1) g^{(k-1)}(x) = 0$$

$$\text{则 } g^{(k+1)}(0) + k(k-1) g^{(k-1)}(0) = 0 \quad (k \geq 2)$$

$$\text{而 } g(0) = g'(0) = 0, g''(0) = 2$$

$$\therefore g^{(k)}(0) = \begin{cases} 0, & k=0 \text{ 或 } k \text{ 为奇} \\ 2 \cdot (k-1)!, & k \text{ 为偶} \end{cases}$$

$$\therefore f^{(n)}(0) = \begin{cases} 0, & n=1 \text{ 或 } n \text{ 为偶} \\ 2n \cdot (n-2)!, & n \text{ 为奇} \end{cases}$$

$$4. \int_0^t \sqrt{1+f'(x)} dx = \int_0^t f(x) dx, \forall t > 0$$

$$\therefore \sqrt{1+f'(x)} \equiv f(x)$$

$$1 + \frac{df}{dx} = f^2$$

$$\frac{df}{(f-1)(f+1)} = dx$$

$$x = \int \frac{df}{f-1} - \frac{1}{2} \int \frac{df}{f+1} = \frac{1}{2} [\ln(f-1) - \ln(f+1)] + C$$

$$\therefore f(x) = \frac{2}{1 - e^{2x-2C}} - 1$$

$$\text{由 } f(0) = 1, \text{ 得 } C = +\infty, \text{ 则 } f(x) \equiv 1$$

$$\therefore V = \pi t$$

$$5. (1) \text{ 设 } z = y' \cos x, \text{ 则 } \sin x \cdot z' - \cos x \cdot z = \sin^3 x + 1$$

$$\sin x \cdot z' - \cos x \cdot z = 0 \Rightarrow z = C \sin x$$

$$dy = C_1 \sin x dx \Rightarrow y = -C_1 \cos x + C_2$$

$$(2) \text{ 设 } z = y' - y, \text{ 则 } z' - 2z = 2x \Rightarrow z = -x - \frac{1}{2} + C e^{2x}$$

$$(y e^{-x})' = -e^{-x} (x + \frac{1}{2}) \Rightarrow y = x + \frac{3}{2} + C_1 e^{2x} + C_2 e^x$$

33 2016-2017 第一学期期末考试

$$1. (1) = \lim_{x \rightarrow 0^+} \frac{\sin x \cdot 2x}{3x^2} = \lim_{x \rightarrow 0^+} \frac{2x^2}{3x^2} = \frac{2}{3}$$

$$(2) = \lim_{n \rightarrow \infty} n(1-\zeta) \cos(n\zeta) \sin \zeta^n \quad (\zeta \in (0, 1))$$

$$= \lim_{n \rightarrow \infty} n(1-\zeta) \cos(n\zeta) \zeta^n = 0$$

$$2. (1) = \frac{1}{2} \int \arctan x dx^2 = \frac{1}{2} x^2 \arctan x - \frac{1}{2} \int x^2 d \arctan x$$

$$= \frac{1}{2} x^2 \arctan x - \frac{1}{2} \int \frac{x^2}{1+x^2} dx$$

$$= \frac{1}{2} x^2 \arctan x - \frac{1}{2} x + \frac{1}{2} \int \frac{1}{1+x^2} dx$$

$$= \frac{1}{2} x^2 \arctan x - \frac{1}{2} x + \frac{1}{2} \arctan x + C$$

$$(2) = \int_1^{+\infty} \left(\frac{1}{x} - \frac{1}{1+x} \right) dx = \ln \frac{x}{1+x} \Big|_1^{+\infty} = \ln 2$$

$$(3) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \cos x d \sin x = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin x \cos x)^2 dx = \sqrt{}$$

3. 在 $[\delta, +\infty)$ 上一致收敛

$$|u_n| \leq \frac{1}{n^2 \delta}$$

在 $(0, +\infty)$ 上不一致收敛

$$\text{取 } x = \frac{1}{n^2}, \lim_{n \rightarrow \infty} \sup_{x \in (0, +\infty)} u_n(x) \geq \lim_{n \rightarrow \infty} u_n\left(\frac{1}{n^2}\right) = \lim_{n \rightarrow \infty} \frac{\cos \frac{1}{n}}{2} = \frac{1}{2} > 0$$

$$4. \lim_{n \rightarrow \infty} \frac{a_n}{a_{n+1}} = 2 \Rightarrow R = 2$$

$$\text{设 } y = \frac{x}{2}, \text{ 则级数 } = \sum_{n=1}^{\infty} \frac{y^n}{n} = f(y)$$

$$\frac{f(y)}{y} = \sum_{n=1}^{\infty} \frac{y^{n-1}}{n} = \left(\sum_{n=1}^{\infty} y^n \right)' = \left(\frac{1}{1-y} \right)' = \frac{1}{(1-y)^2}$$

$$\therefore \text{级数的和函数} = \frac{y}{(1-y)^2} = \frac{2x}{(2-x)^2}$$

$$6. LHS = \int_0^a x f(x) dx + \int_a^{2a} x f(x) dx$$

$$= \int_0^a x f(x) dx + \int_0^a (x+a) f(x+a) dx$$

$$= \int_0^a x f(x) dx + \int_0^a -(x+a) f(x) dx$$

$$= -a \int_0^a f(x) dx$$

$$7. \Leftarrow 2 \int_0^x x f(x) dx \geq \left(\int_0^x f(x) dx \right)^2, \forall x \in [0, 1]$$

$$\Leftarrow (\text{求导}) 2x f(x) \geq 2 f(x) \int_0^x f(x) dx$$

$$\Leftarrow x \geq \int_0^x f(x) dx$$

$$\Leftarrow \int_0^x (1-f(x)) dx \geq 0$$

"=" eg. $f(x) \equiv 0$ 或 $f(x) \equiv 1$

Memo NO. _____

Date / /

X-Talk

5. (1) $y^{-2}y' + y^{-1} = x$

设 $z = y^{-1}$, 则 $-z' + z = x$

将 $z = ue^x$ 代入, 得 $-u'e^x = x \Rightarrow u = -\int \frac{x}{e^x} dx = e^{-x}(1+x) + C$

$\therefore y = \frac{1}{z} = \frac{1}{1+x+Ce^x}$

显然 $y=0$ 也成立

(2) 设 $z = y'$, 则 $xz' + 2z = 2$

$\frac{dz}{2(1-z)} = \frac{dx}{x}$

$-\frac{1}{2}\ln(1-z) = \ln x + \ln C_1 \Rightarrow z = 1 - \frac{C}{x^2}$

~~$z = 1 - \frac{C}{x^2}$~~ $y = \int (1 - \frac{C}{x^2}) dx = x + \frac{C}{x} + C'$

~~$y' = 1 - \frac{C}{x^2} = z$~~

~~$y = x - \frac{1}{3}Cx^3 + C' = x - \frac{1}{3}Cx^3 + C'$~~

6. 设 $z = y'$, 则 $z' - 2z = e^{4x}$

将 $z = ue^{2x}$ 代入, 得 $u'e^{2x} = e^{4x} \Rightarrow u = \int e^{2x} dx = \frac{1}{2}e^{2x} + C$

$\therefore y' = z = \frac{1}{2}e^{4x} + Ce^{2x}$ $y'(0) = 0 \Rightarrow C = -\frac{1}{2}$

$y = \frac{1}{8}e^{4x} + C'e^{2x} + C''$ $y(0) = 0 \Rightarrow C'' = \frac{1}{8}$

$\therefore y = \frac{1}{8}e^{4x} - \frac{1}{4}e^{2x} + \frac{1}{8}$

7. (1) 由数学归纳法.

$\frac{f_{n+1}(x)}{f_n(x)} = \frac{2}{1 + \frac{f_n(x)}{f_n(x)}} \leq \frac{2}{1 + \frac{1}{C_n}} = \frac{2C_n}{C_n+1} = C_{n+1}$

(2) ~~$\frac{f_n(x)}{f_{n-1}(x)} = \frac{f_{n-1}(x)}{f_{n-2}(x)}$~~

若 $f_1 \geq f_0$ ($f_1 < f_0$ 同理可证), 则 $f_{n+1} \geq f_n$
而 $f_n \leq f_{n-1} \leq 2f_{n-1}$, 故 $(f_n - 2f_{n-1})(f_n - f_{n-1}) \leq 0$

Memo NO. _____

Date / /

X-Talk

$6f_n^2 - 5f_n(f_n + f_{n-1}) + 2f_{n-1}(f_n + f_{n-1}) \leq 0$

$3f_{n+1} - 5f_n + 2f_{n-1} \leq 0$

$f_{n+1} - f_n \leq \frac{2}{3}(f_n - f_{n-1})$

$\therefore |f_{n+1} - f_n| \leq (\frac{2}{3})^n |f_1 - f_0|$

(3) 由柯西收敛准则. 设 $M = \max |f_1(x) - f_0(x)|$

$\forall \varepsilon > 0, \exists N = [\log_{\frac{2}{3}} \frac{\varepsilon}{3M}], \forall n > N, p > 0$, 有

$|f_n(x) - f_{n+p}(x)| \leq |f_n(x) - f_{n+1}(x)| + \dots + |f_{n+p-1}(x) - f_{n+p}(x)|$
 $\leq [(\frac{2}{3})^{n+1} + \dots + (\frac{2}{3})^{n+p}] |f_1(x) - f_0(x)|$
 $\leq (\frac{2}{3})^{n+1} \cdot \frac{1}{1-\frac{2}{3}} \cdot M = 3M(\frac{2}{3})^{n+1} < \varepsilon$

8. 充分性: $\{a_n\}$ 单调有界 $\Rightarrow$ 收敛 $\Rightarrow \sum_{n=1}^{\infty} (a_{n+1} - a_n)$ 收敛

$a_n > 1 \Rightarrow \frac{1}{a_n \ln a_{n+1}}$ 单调有界 $\Rightarrow \sum_{n=1}^{\infty} \frac{a_{n+1} - a_n}{a_n \ln a_{n+1}}$ 收敛
(或: $\sum_{n=1}^{\infty} \frac{a_{n+1} - a_n}{a_n \ln a_{n+1}} \leq \sum_{n=1}^{\infty} \frac{a_{n+1} - a_n}{a_1 \ln a_2} = \frac{a_{\infty} - a_1}{a_1 \ln a_2}$)

必要性: (若题目改为 $\sum_{n=1}^{\infty} \frac{a_{n+1} - a_n}{a_n \ln a_n}$ 则:

反证法: 若 $\{a_n\}$ 无界, 则 $\frac{\ln \ln a_{n+1} - \ln \ln a_n}{a_{n+1} - a_n} = \frac{1}{a_{n+1} \ln a_{n+1}} < \frac{1}{a_n \ln a_n}$
级数 $> \sum_{n=1}^{\infty} (\ln \ln a_{n+1} - \ln \ln a_n)$ 无界 $\Rightarrow$ 发散)

Memo NO. _____

Date / /

X-Talk

34] 2017-2018 第一学期期末考试

1. $P_n(x) = \sum_{k=0}^n a_k (x-x_0)^k$

2. (1) $\int \frac{dx}{2 \sin \frac{x}{2} \cos \frac{x}{2}} = \int \frac{\sec^2 \frac{x}{2}}{2 \tan \frac{x}{2}} dx = \int \frac{d \tan \frac{x}{2}}{\tan \frac{x}{2}} = \ln |\tan \frac{x}{2}| + C$

(2) $\int e^t dt^3 = 3 \int t^2 e^t dt = 3(t^2 e^t - \int e^t dt^2)$
 $= 3(t^2 e^t - 2 \int t e^t dt) = 3(t^2 e^t - 2te^t + 2 \int e^t dt)$
 $= 3(t^2 e^t - 2te^t + 2e^t) + C = 3(x^{\frac{2}{3}} e^{\sqrt[3]{x}} - 2\sqrt[3]{x} e^{\sqrt[3]{x}} + 2e^{\sqrt[3]{x}})$

(3) $\int_0^1 \arctan x dx - \int_0^1 \arctan x d \arctan x$
 $= x \arctan x |_0^1 - \int_0^1 \frac{x}{1+x^2} dx - \frac{1}{2} (\arctan x)^2 |_0^1$
 $= \frac{\pi}{4} - \frac{1}{2} \ln(1+x^2) |_0^1 - \frac{1}{2} (\frac{\pi}{4})^2 = \frac{\pi}{4} - \frac{1}{2} \ln 2 - \frac{\pi^2}{32}$
 (4) $\int_{-\frac{\pi}{2}}^0 \frac{\sin^2 x}{1+e^x} dx + \int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{1+e^x} dx = \int_0^{\frac{\pi}{2}} \sin^2 x (\frac{1}{1+e^x} + \frac{1}{1+e^{-x}}) dx$
 $= \int_0^{\frac{\pi}{2}} \sin^2 x dx = \int_0^{\frac{\pi}{2}} \frac{1-\cos 2x}{2} dx = (\frac{x}{2} - \frac{\sin 2x}{4}) |_0^{\frac{\pi}{2}} = \frac{\pi}{4}$

3. (1) $\lambda^3 + \lambda^2 + \lambda + 1 = 0 \Rightarrow \lambda = -1, \pm i$
 $\therefore y = C_0 e^{-x} + C_1 \cos x + C_2 \sin x$

(2) $y' + 2xy = 0 \Rightarrow y = C e^{-x^2}$
 将 $y = u e^{-x^2}$ 代入, 得 $u' e^{-x^2} = 4x \Rightarrow u = \int \frac{4x}{e^{x^2}} dx = 2e^{x^2} + C$
 $\therefore y = 2 + C e^{-x^2}$
 由 $y(0) = 0 \Rightarrow C = -2 \Rightarrow y = 2 - 2e^{-x^2}$

1/4. $\Leftarrow x \int_0^x f(t) d \sin^2 \frac{t}{2} \geq \sin^2 \frac{x}{2} \int_0^x f(t) dt$
 $\Leftarrow$ (分部积分) $x f(x) \sin^2 \frac{x}{2} - x \int_0^x \sin^2 \frac{t}{2} df(t) \geq x f(x) \sin^2 \frac{x}{2} - \sin^2 \frac{x}{2} \int_0^x t df(t)$

$\Leftarrow \int_0^x \frac{t}{x} df(t) \geq \int_0^x \frac{\sin^2 \frac{t}{2}}{\sin^2 \frac{x}{2}} df(t) \Leftarrow \int_0^x (\frac{t}{x} - \frac{\sin^2 \frac{t}{2}}{\sin^2 \frac{x}{2}}) f'(t) dt \geq 0$ (若 $f(x)$ 可导即证)

Memo NO. _____

Date / /

X-Talk

(不可导的证法)

$\Leftarrow x \int_0^x f(t) d \sin^2 \frac{t}{2} \geq \sin^2 \frac{x}{2} \int_0^x f(t) dt$
 $\Leftarrow \int_0^x f(t) d \frac{\sin^2 \frac{t}{2}}{\sin^2 \frac{x}{2}} \geq \int_0^x f(t) d \frac{t}{x}$

(设 $g(t) = \frac{\sin^2 \frac{t}{2}}{\sin^2 \frac{x}{2}}, h(t) = \frac{t}{x}$)
 $\Leftarrow g(t) \leq h(t), \forall t \in [0, x]$
 $\Leftarrow g(t)$ 为下凸函数

5. (1) $\sum_{n=1}^{\infty} a_n a_n = (-1)^n$ 部分和有界

$b_n = x e^{-nx}$ 单调 $\downarrow$

$x > 0$ 时, $b_n \leq \frac{x}{nx} = \frac{1}{n} \rightarrow 0$; $x = 0$ 时, $b_n = 0$

$\therefore$ 一致收敛

(2) $|u_n(x)| = \frac{x}{(e^x)^n}$

$x > 0$ 时 $\because e^x > 1$

$\therefore$ 收敛

$x = 0$ 时, 级数 $= 0$ $\therefore$ 收敛

(3) $\limsup_{n \rightarrow \infty} |u_n(x)| \geq \lim_{n \rightarrow \infty} |u_n(\frac{1}{n^2})|$

若其一致收敛, 则 $\sum_{n=1}^{\infty} |u_n(x)| = \frac{x}{e^x} \cdot \frac{1}{1 - \frac{1}{e^x}} = \frac{x}{e^x - 1} \triangleq f(x), x \in (0, 1]$

由连续性知 $\lim_{x \rightarrow 0} f(x) \stackrel{L'H}{=} \lim_{x \rightarrow 0} \frac{1}{e^x - 1} = 1$

而 $\sum_{n=1}^{\infty} |u_n(0)| = 0$, 矛盾

$\therefore$ 不一致收敛

6. 设 $x-2=3y$, 则 $f(x) = \ln(3y+3) = \ln 3 + \ln(1+y)$
 $= \ln 3 + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{y^n}{n} \quad (-1 < y \leq 1)$
 $= \ln 3 + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(x-2)^n}{n \cdot 3^n} \quad (-1 < x \leq 5)$

7. $g(x)$ 在 x_0 可导: $g'(x_0) = \lim_{x \rightarrow x_0} \frac{g(x) - g(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} \frac{1}{x - x_0} [f(x) - f(x_0) - f'(x_0)(x - x_0)]$

见反面

Date / /

X-Talk

8. 设 $|f'(x_1)| = M = \max |f'(x)|$

$$|f'(x_2)| = m = \min |f'(x)|$$

$$\text{则 } \int_0^1 |f'(x)| dx \leq M$$

$$\int_0^1 |f''(x)| dx \geq \int_{x_2}^{x_1} |f''(x)| dx \geq \left| \int_{x_2}^{x_1} f''(x) dx \right| = |f'(x_1) - f'(x_2)| = M - m$$

$$f(0)f(1) \geq 0 \Rightarrow \text{不妨设 } f(0), f(1) \geq 0 \quad \therefore \text{只需证 } \int_0^1 |f(x)| dx \geq \frac{m}{2}$$

若 $\exists x \in [0, 1]$ s.t. $f'(x_0) = 0$, 则 $m = 0$, 显然成立

若不存在, 不妨设 $\forall x \in [0, 1], f'(x) > 0 \Rightarrow m = \min f'(x)$

$$\text{则 } f(x) \geq f(0) + \int_0^x f'(x) dx \geq f(0) + mx$$

$$\int_0^1 |f(x)| dx \geq \int_0^1 mx dx = m \cdot \frac{1}{2} x^2 \Big|_0^1 = \frac{m}{2} \quad \therefore \text{得证}$$

$$7. g'(x_0) = \lim_{h \rightarrow 0} \frac{g(x_0+h) - g(x_0)}{h} = \lim_{h \rightarrow 0} \frac{\frac{f(x_0+h) - f(x_0)}{h} - f'(x_0)}{h} = \lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0) - f'(x_0)h}{h^2}$$

$$\stackrel{L'H}{=} \lim_{h \rightarrow 0} \frac{f'(x_0+h) - f'(x_0)}{2h}$$

$$\lim_{x \rightarrow x_0} g'(x) = \lim_{x \rightarrow x_0} \lim_{h \rightarrow 0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} \frac{f'(x)(x - x_0) - f(x) + f(x_0)}{(x - x_0)^2} = \lim_{h \rightarrow 0} \frac{f'(x_0+h)h - f(x_0+h) + f(x_0)}{h^2}$$

$$\therefore g'(x_0) - \lim_{x \rightarrow x_0} g'(x) = \lim_{h \rightarrow 0} \frac{2(f(x_0+h) - f(x_0)) - (f'(x_0) + f'(x_0+h))h}{h^2}$$

$$= \lim_{h \rightarrow 0} \frac{2f(x_0+h) - f'(x_0)h - f'(x_0+h)h}{h^2} = \lim_{h \rightarrow 0} \frac{f'(x_0)h - f'(x_0+h)h}{h^2} = -f''(x_0)$$

$$\stackrel{L'H}{=} \lim_{h \rightarrow 0} \frac{f'(x_0+h)h - f'(x_0)h}{h^2}$$

$$\therefore g'(x_0) - \lim_{x \rightarrow x_0} g'(x) = \lim_{h \rightarrow 0} \frac{f'(x_0+h)h - f'(x_0)h - 2f(x_0+h)h + 2f(x_0+h) - 2f(x_0)}{2h^2}$$

$$= \lim_{h \rightarrow 0} \frac{-(f'(x_0+h) - f'(x_0))h - 2f(x_0)h + 2f(x_0+h) - 2f(x_0)}{2h^2} = -\frac{f''(x_0)}{2} + \lim_{h \rightarrow 0} \frac{-f'(x_0)h + f(x_0+h) - f(x_0)}{h^2}$$

$$\stackrel{L'H}{=} -\frac{f''(x_0)}{2} + \lim_{h \rightarrow 0} \frac{-f'(x_0) + f'(x_0+h)}{2h} = -\frac{f''(x_0)}{2} + \frac{f''(x_0)}{2} = 0$$

Memo NO. _____

Date / /

X-Talk

35 2018-2019 第一学期期末考试

$$1. (1-x)f(x) = e^x, f(0) = 1$$

$$\text{两边求 } n \text{ 次导, } (1-x)f^{(n)}(x) - nf^{(n-1)}(x) = e^x$$

$$f^{(n)}(0) - nf^{(n-1)}(0) = 1$$

$$\therefore f^{(n)}(0) = nf^{(n-1)}(0) + 1 = 1 + n(1 + (n-1)f^{(n-2)}(0))$$

$$= \dots = 1 + n + n(n-1) + \dots + \frac{n!}{1!} + n!f^{(0)}(0)$$

$$= \sum_{k=0}^n \frac{n!}{k!}$$

$$2. (1) = \int dx - \int \frac{e^{2x}}{1+e^{2x}} dx = x - \frac{1}{2} \int d(1+e^{2x}) = x - \frac{1}{2} \ln(1+e^{2x}) + C$$

$$(2) = \int \frac{dx}{(x-1)(x+1)^2} = \frac{1}{4} \int \left(\frac{1}{x-1} - \frac{x+3}{(x+1)^2} \right) dx$$

$$= \frac{1}{4} \int \left(\frac{1}{x-1} - \frac{1}{x+1} - \frac{2}{(x+1)^2} \right) dx$$

$$= \frac{1}{4} [\ln(x-1) - \ln(x+1) + \frac{2}{x+1}] + C$$

$$(3) = \frac{1}{2} \int_0^1 \arctan x dx^2 = \frac{1}{2} x^2 \arctan x \Big|_0^1 - \frac{1}{2} \int_0^1 \frac{x^2}{1+x^2} dx$$

$$= \frac{\pi}{8} - \frac{1}{2} \left(\int_0^1 dx - \int_0^1 \frac{1}{1+x^2} dx \right)$$

$$= \frac{\pi}{8} - \frac{1}{2} (x - \arctan x) \Big|_0^1 = \frac{\pi}{4} - \frac{1}{2}$$

$$(4) \text{ 令 } t = \sqrt{x-1}, x = t^2 + 1$$

$$= \int_0^{+\infty} \frac{2t dt}{(t^2+1)t} = 2 \int_0^{+\infty} \frac{dt}{t^2+1} = 2 \arctan t \Big|_0^{+\infty} = \pi$$

$$3. (1) ((1+x^2)y')' = x$$

$$(1+x^2)y' = \frac{1}{2}x^2 + C_1$$

$$y' = \frac{1}{2} \cdot \frac{x^2+C_1}{x^2+1} = \frac{1}{2} \left(1 + \frac{C_1}{x^2+1} \right)$$

$$y = \frac{1}{2}x + C_1 \arctan x + C_2$$

Memo NO. _____

Date / /

X-Talk

(2) $z = y' - y$, 则 $z' - 2z = 2x - 3$ ✓

4. $LHS = \int_0^{\frac{\pi}{2}} x f(|\cos x|) dx + \int_{\frac{\pi}{2}}^{\pi} x f(|\cos x|) dx$

$$= \int_0^{\frac{\pi}{2}} x f(|\cos x|) dx + \int_0^{\frac{\pi}{2}} (\pi - x) f(|\cos x|) dx = RHS$$

5. 一致收敛.

令 $x = 1 + \alpha$.
由微分中值定理 $\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha} = \frac{\alpha}{(n-\theta)^{\alpha+1}} > \frac{\alpha}{n^\alpha}$

$$\sum_{n=1}^{\infty} \frac{(x-1)^2}{n^\alpha} < (x-1)^2 + \sum_{n=2}^{\infty} \left[\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha} \right] = (x-1)^2 + 1$$

$\forall \varepsilon > 0, \exists N = [e^{\frac{1}{\varepsilon}}] + 1, \forall n > N, p > 0, x > 1$, 有

$$\frac{(x-1)^2}{(n+1)^\alpha} + \frac{(x-1)^2}{(n+2)^\alpha} + \dots + \frac{(x-1)^2}{(n+p)^\alpha} < \alpha \left(\frac{1}{n^\alpha} - \frac{1}{(n+p)^\alpha} \right) < \frac{\alpha}{n^\alpha} < \frac{\frac{1}{\ln n}}{n^{\frac{1}{\ln n}}} < \frac{1}{\ln n} < \varepsilon$$

6. ✓

7. (1) $\frac{a_n}{a_{n+1}} \leq \frac{n \ln n + \ln n + 1}{n \ln n} + \frac{1}{n^2} < \frac{(n+1) \ln(n+1)}{n \ln n} + \frac{(n+1) \ln(n+1)}{n \ln n} \cdot \frac{1}{n^2}$

(2) $\ln \frac{a_n}{a_{n+1}} \leq \ln \left(1 + \frac{1}{n} + \frac{1}{n \ln n} + \frac{1}{n^2} \right) \leq \frac{1}{n} + \frac{1}{n \ln n} + \frac{1}{n^2}$

累加 $\Rightarrow \ln a_3 - \ln a_n \leq \int_2^n \frac{1}{x} dx + \int_2^n \frac{1}{x \ln x} dx + C = \ln(x \ln x) \Big|_{x=2}^n + C$
 $= \ln(n \ln n) + C'$

$\Rightarrow a_n \geq \frac{C''}{n \ln n} > 0$ 而 $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ 发散, 故 $\sum_{n=2}^{\infty} a_n$ 发散

8. 对于任意给定的 x :

若 $f'(x) = 0$, 则已成立

若 $f'(x) \neq 0$, 不妨设 $f'(x) > 0$ (< 0 同理)

取 $h = (f'(x))^{\frac{1}{\alpha}}$, 则 $\int_{x-h}^x (f'(x) - f'(t)) dt = -f(x) + f(x-h) + f'(x)h$

$$\int_{x-h}^x (x-t)^\alpha dt \geq -f(x) + (f'(x))^{\frac{\alpha+1}{\alpha}}$$

$$\int_0^h x^\alpha dx = \frac{x^{\alpha+1}}{\alpha+1} \Big|_0^h = \frac{(f'(x))^{\frac{\alpha+1}{\alpha}}}{\alpha+1}$$

此题取 $\alpha = \frac{1}{2}$ 即可

Memo NO. _____

Date / /

X-Talk

36 2019-2020 第一学期期末考试

1. (1) $\frac{1}{2} \int \left(-\frac{1}{x^2-1} + \frac{1}{x^2+1} \right) dx = \frac{1}{4} \int \left(\frac{1}{x-1} - \frac{1}{x+1} \right) + \frac{1}{2} \int \frac{1}{x^2+1} dx$

$$= -\frac{1}{4} [\ln(x-1) - \ln(x+1)] + \frac{1}{2} \arctan x + C$$

(2) $t = \sqrt{x}, x = t^2$

$$= \int_0^2 \frac{2t}{1+t} dt = \int_0^2 \left(2 - \frac{2}{1+t} \right) dt = [2t - 2 \ln(1+t)] \Big|_0^2 = 4 - 2 \ln 3$$

(3) $I = \int_0^{+\infty} \cos x de^x = -\cos x \cdot e^x \Big|_0^{+\infty} + \int_0^{+\infty} e^x d \cos x$

$$= 1 + \int_0^{+\infty} \sin x de^x = 1 + \sin x \cdot e^x \Big|_0^{+\infty} - \int_0^{+\infty} e^x \cos x dx = 1 - I$$

$$\therefore I = \frac{1}{2}$$

(4) 当 $x \geq 1$ 时 $\int \ln x dx = x(\ln x - 1) + C_1$

当 $x < 1$ 时 $= -\int \ln x dx = -x(\ln x - 1) + C_2 = -x(\ln x - 1) + C_1 - 2$

(5) $= \lim_{n \rightarrow \infty} e^{(1^p + 2^p + \dots + n^p) \cdot \frac{1}{n^{p+1}}} = \exp \left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \left(\frac{i}{n} \right)^p \right)$

$$= \exp \left(\int_0^1 x^p dx \right) = \exp \left(\frac{x^{p+1}}{p+1} \Big|_0^1 \right) = e^{\frac{1}{p+1}}$$

2. $y'(0) = 2, y(0) = 0$

设 $z = y' - 3y$, 则 $z(0) = 2$

$$z' - 3z = e^{3x} \quad (*)$$

$$z' - 3z = 0 \Rightarrow z = Ce^{3x}$$

将 $z = ue^{3x}$ 代入 (*) 得 $u'e^{3x} = e^{3x} \Rightarrow u = x + C$

$$\therefore z = (x + C)e^{3x}, z(0) = C = 2$$

$$\therefore y' - 3y = (x + 2)e^{3x}$$

$$y = e^{3x} \left(\int (x + 2) dx + C \right) = e^{3x} \left(\frac{1}{2} x^2 + 2x + C \right)$$

$$\therefore y = x \left(\frac{1}{2} x + 2 \right) e^{3x}$$

Date / /

X-Talk

3. $x, y > 0$

当 $x, y > 1$ 时 $\ln x + \ln y = 1 \Rightarrow xy = e$

$$S_1 = \int_1^e \left(\frac{e}{x} - 1 \right) dx = (e \ln x - x) \Big|_1^e = 1$$

当 $x > 1, y < 1$ 时 $\ln x - \ln y = 1 \Rightarrow y = \frac{x}{e}$

$$S_2 = \int_1^e \frac{1}{2} \left(1 - \frac{1}{e} \right) (e - 1) = \frac{(e-1)^2}{2e}$$

当 $x < 1, y > 1$ 时 $S_3 = S_2$

当 $x, y < 1$ 时 $-\ln x - \ln y = 1 \Rightarrow xy = \frac{1}{e}$

$$S_4 = \int_{\frac{1}{e}}^1 \left(1 - \frac{1}{ex} \right) dx = \left(x - \frac{1}{e} \ln x \right) \Big|_{\frac{1}{e}}^1 = 1 - \frac{2}{e}$$

$$\therefore S = 1 + \frac{(e-1)^2}{e} + 1 - \frac{2}{e} = e - \frac{1}{e}$$

4. $\alpha > 0$ 或 $(\beta < 0 \text{ 且 } \alpha > \beta)$ 时: $f(0^+) = 0$ 连续 $\Rightarrow$ 可积

$\alpha = 0$ 时: 有界 + 至多有限个间断点 $\Rightarrow$ 可积

否则: 无界 $\Rightarrow$ 不可积

$$5. (1) \frac{1}{(k-1)^{\alpha-1}} - \frac{1}{k^{\alpha-1}} = \frac{\alpha-1}{(k-\theta)^{\alpha}} > \frac{\alpha-1}{k^{\alpha}} \quad \text{见 [35] T5}$$

$$\therefore \sum_{k=1}^n \frac{1}{k^{\alpha}} < \left(1 + 1 - \frac{1}{n^{\alpha-1}} \right) \cdot \frac{1}{\alpha-1} = \frac{1}{\alpha-1} \left(2 - \frac{1}{n^{\alpha-1}} \right) \quad (n \geq 2)$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \cdot \frac{1}{\alpha-1} \left(2 - \frac{1}{n^{\alpha-1}} \right) = \sum_{n=1}^{\infty} \frac{1}{\alpha-1} \left(\frac{2}{n^2} - \frac{1}{n^{\alpha+1}} \right) \text{ 收敛 } \therefore \text{收敛}$$

$$(2) \text{ 当 } n \rightarrow \infty \text{ 时, } \frac{x}{n} - \sin \frac{x}{n} \sim \frac{x}{n} - \frac{x}{n} + o\left(\left(\frac{x}{n}\right)^3\right) = o\left(\left(\frac{x}{n}\right)^3\right)$$

$$\text{而 } \sum_{n=1}^{\infty} \frac{x^3}{n^3} \text{ 一致收敛 } \therefore \text{一致收敛}$$

6. 设 $x = f(t), t = f^{-1}(x)$

$$\begin{aligned} \text{则 } \int f^{-1}(x) dx &= \int t dx = tx - \int x dt = tx - \int f(t) dt \\ &= tx - F(t) + C = xf^{-1}(x) - F(f^{-1}(x)) + C \end{aligned}$$

Memo NO. _____

Date / /

X-Talk

7. 必要性: $\int_0^T f(x) dx = F(T) - F(0) = 0$

充分性: $0 = \int_0^{x_0} f(x) dx + \int_{x_0}^T f(x) dx$

$$= \int_0^{x_0} f(x+T) dx + \int_{x_0}^T f(x) dx$$

$$= \int_{x_0}^{T+x_0} f(x) dx = F(x_0+T) - F(x_0), \forall x_0$$

$$= f(T+x_0) \therefore \text{以 } T \text{ 为周期}$$

8. $x=1$ 时发散.

由 Abel 定理知 $|x| > 1$ 时也发散.

$|x| < 1$ 时: 设 $|a_n| \leq M, \forall n$

对于给定的 $x \in (-1, 1), \forall \varepsilon > 0, \exists N = \left\lceil \log_x \frac{\varepsilon(1-x)}{M} \right\rceil, \forall n > N, p > 0$, 有

$$\left| a_{n+1}x^{n+1} + a_{n+2}x^{n+2} + \dots + a_{n+p}x^{n+p} \right|$$

$$\leq Mx^{n+1} + Mx^{n+2} + \dots + Mx^{n+p}$$

$$= Mx^{n+1} \cdot \frac{1-x^p}{1-x} < \frac{Mx^{n+1}}{1-x} < \varepsilon$$

由 Cauchy 收敛准则知其收敛

Memo NO. _____

Date / /

X-Talk

37 2020-2021 第一学期期末考试

$$1. (1) = \int \frac{\sin x (1 - \cos^2 x)}{\cos x} dx = \int (\tan x - \frac{1}{2} \sin 2x) dx = \ln |\cos x| + \frac{1}{4} \cos 2x + C$$

$$(2) = x(\ln x - 1) \Big|_0^1 = -1$$

$$(3) = \lim_{x \rightarrow 0} \frac{\int_0^x t^2 dt}{x^3} = \lim_{x \rightarrow 0} \frac{x^2}{3x^2} = \frac{1}{3}$$

$$(4) \blacksquare y' - \frac{2}{x}y = 0 \Rightarrow \frac{dy}{y} = \frac{2dx}{x} \Rightarrow \ln y = 2\ln x + C \Rightarrow y = Cx^2$$

将 $y = ux^2$ 代入, 得 $u'x^2 = 2x^2 \Rightarrow u = 2x + C$

$$\therefore y = 2x^3 + Cx^2$$

$$(5) \sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad (-1 < x < 1)$$

求导, 得 $\sum_{n=1}^{\infty} nx^{n-1} = \frac{1}{(x-1)^2}$ 即 $\sum_{n=1}^{\infty} nx^n = \frac{x}{(x-1)^2}$

$x = \pm 1$ 时不收敛

$$\therefore (-1, 1) \quad S(x) = \frac{x}{(x-1)^2}$$

$$2. r = 1 + \cos \theta$$

$$S = 2\pi \int_0^\pi r \sin \theta \sqrt{r^2 + r'^2} d\theta = 2\pi \int_0^\pi (1 + \cos \theta) \sin \theta \sqrt{2 + 2\cos \theta} d\theta$$

$$= 2\pi \int_0^\pi 2\cos^2 \frac{\theta}{2} \cdot 2\sin \frac{\theta}{2} \cos \frac{\theta}{2} \cdot 2\cos \frac{\theta}{2} d\theta$$

$$= 32\pi \int_\pi^0 \cos^4 \frac{\theta}{2} d\cos \frac{\theta}{2} = 32\pi \cdot \frac{1}{5} \cos^5 \frac{\theta}{2} \Big|_\pi^0 = \frac{32}{5}\pi$$

$$3. \text{ 设 } z = y' - 2iy, \text{ 则 } z' + 2iz = 9x \sin x$$

$$(e^{2ix} z)' = 9x \sin x \cdot e^{2ix}$$

题中方程为 $y'' + 4y = 9x e^{ix}$ 的虚部

$$y = z \cdot e^{ix} \Rightarrow z'' + 2iz' + 3z = 9x$$

待定系数法得特解 $\tilde{z} = 3x - 2i \Rightarrow \tilde{y}_0 = (3x - 2i)e^{ix}$

分离出虚部, 得 $\tilde{y}_0 = 3x \sin x - 2\cos x$

对应齐次方程 $y'' + 4y = 0$ 的基本解组为 $\{\cos 2x, \sin 2x\}$

$$\therefore \text{通解 } y = C_1 \cos 2x + C_2 \sin 2x + 3x \sin x - 2\cos x$$

Memo NO. _____

Date / /

X-Talk

$$4. I = \int_0^{\frac{\pi}{2}} \frac{x \sin x}{\sin x + \cos x} dx + \int_{\frac{\pi}{2}}^{\pi} \frac{x \sin x}{\sin x - \cos x} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{x \sin x}{\sin x + \cos x} dx + \int_0^{\frac{\pi}{2}} \frac{(\pi - x) \sin x}{\sin x + \cos x} dx = \int_0^{\frac{\pi}{2}} \frac{\pi \sin x}{\sin x + \cos x} dx$$

$$xI = \int_0^{\frac{\pi}{2}} \frac{\pi \sin(\frac{\pi}{2} - x)}{\sin(\frac{\pi}{2} - x) + \cos(\frac{\pi}{2} - x)} dx = \int_0^{\frac{\pi}{2}} \frac{\pi \cos x}{\sin x + \cos x} dx$$

$$\therefore 2I = \int_0^{\frac{\pi}{2}} \pi dx = \frac{\pi^2}{2} \Rightarrow I = \frac{\pi^2}{4}$$

$$5. f(x) = \frac{1}{x+1} - \frac{1}{x+2} = \frac{1}{(x+4)-3} - \frac{1}{(x+4)-2}$$

$$= \frac{1}{3} \cdot \frac{1}{\frac{x+4}{3}-1} - \frac{1}{2} \cdot \frac{1}{\frac{x+4}{2}-1}$$

$$= -\frac{1}{3} \sum_{n=0}^{\infty} \left(\frac{x+4}{3}\right)^n + \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{x+4}{2}\right)^n$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2^{n+1}} - \frac{1}{3^{n+1}}\right) (x+4)^n \quad (-6, -2)$$

$$6. \cancel{e^{nx} \geq nx+1 \geq nx} \Rightarrow \cancel{2^{nx} \geq nx+1 \geq nx} \Rightarrow \cancel{2^{nx} \geq nx}$$

求导可证 逐点收敛显然

不一致收敛: 法I: $x=0$ 处不连续

法II: $f(x) \triangleq \frac{x}{2^{nx}}, f'(x) = \frac{2^{nx} - x \cdot 2^{nx} \cdot n \ln 2}{(2^{nx})^2}$

$$\text{令 } f'(x) = 0 \Rightarrow x = \frac{1}{n \ln 2}$$

$$\therefore \lim_{n \rightarrow \infty} \sup_{x \in J} |u_n| = \lim_{n \rightarrow \infty} u_n\left(\frac{1}{n \ln 2}\right) = 0 \quad \times \text{ (不可行)}$$

7. 单调 $\downarrow \rightarrow 0 \Rightarrow$ 收敛

$$\sum_{k=1}^n a_n = \sum_{k=1}^n \int_0^1 \frac{1}{(1+t^3)^{2k}} dt = \int_0^1 \sum_{k=1}^n \frac{1}{(1+t^3)^{2k}} dt \leq \int_0^1 \frac{1}{(1+t^3)^2} \cdot \frac{1}{1 - \frac{1}{(1+t^3)^2}} dt$$

$$= \int_0^1 \frac{1}{(1+t^3)^2 - 1} dt = \int_0^1 \frac{1}{t^3(t^3+1)} dt \quad \times$$